

SUPPLEMENTARY INFORMATION

Linker guided growth of single-crystal covalent organic frameworks

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Table of Contents

Materials and Methods

Section 1. Synthesis and characterization of multicomponent COFs

Section 2. Continuous rotation electron diffraction (cRED) analysis

Section 3. Powder XRD analysis

Section 4. Solid-state nuclear magnetic resonance spectroscopic analysis

Section 5. Thermogravimetric analysis (TGA)

Section 6. Gas absorption

Section 7. Optical microscopic pictures of synthesis COFs

Section 8. Discussion on adsorption mechanism of sulfur dioxide

Section 9. Discussion of the effect of sequence on SO₂ adsorption

Supplementary Methods

Materials

All solvents and reagents were used as received without further purification unless otherwise specified. *p*-Toluenesulfonic acid monohydrate (purity $\geq 98\%$) and Tetrahydrofuran (AR, purity $\geq 99\%$) were purchased from Aladdin Co. Octanoylglycine (C₈-GlyA, purity $\geq 97\%$) was purchased from Shanghai Bide Pharmaceutical Technology Co., Ltd. Tetrakis(4-aminophenyl)methane (TAM) (purity $\geq 98\%$), 1, 4-phthalaldehyde (purity $\geq 98\%$), 2,5-dihydroxy-1,4-benzenedicarboxaldehyde (purity $\geq 98\%$), 2,5-dimethoxyterephthalaldehyde (purity $\geq 98\%$), 2,5-bis(prop-2-yn-1-yloxy)terephthalaldehyde (purity $\geq 98\%$), 2,3,5,6-tetrafluoroterephthalaldehyde (purity $\geq 98\%$), 2,5-dichloroterephthalaldehyde (purity $\geq 98\%$), 2,5-dibromoterephthalaldehyde (purity $\geq 98\%$), 2-nitroterephthalaldehyde (purity $\geq 98\%$), thieno[3,2-b]thiophene-2,5-dicarboxaldehyde (purity $\geq 98\%$) and [1,1'-biphenyl]-4,4'-dicarbaldehyde (purity $\geq 98\%$) were purchased from Jilin Chinese Academy of Sciences-Yanshen Technology Co., Ltd. Tetrahydrofuran were purchased from Adamas-beta®. Sodium hydroxide (AR, purity $\geq 96\%$) were purchased from Guangzhou Chemical reagent Co.

Instrumentation and Methods

Continuous rotation electron diffraction (cRED)

The cRED data of COF-AB-c, COF-ABDI-a and COF-ABCDEFGHI were collected on a JEOL 2100-plus TEM equipped with MerlinEM direct electron detector under 200kV acceleration voltage. The COF powder was dusted onto a copper grid (R2/2, 200 mesh; QUANTIFOIL) and it was transferred to TEM with a Fischione 2550 cryo-holder. The cRED datasets were collected after the holder tip was cooled to 100K. Each frame was collected with exposure time of 2.0 s, resulting in a 1.0° wedge per frame. The collection time for one dataset was approximately 2 minutes.

Considering the beam sensitivity of COFs, the total electron dose on sample was controlled under 10 e⁻ Å⁻² using low-dose mode during the data collection.

All the data were reconstructed and visualized with program REDp¹ and processed using XDS² with the aid of Instamatic³ for batch processing. The structure solution and refinement were conducted with crystallographic computing system JANA2006⁴. The structural models were visualized with program VESTA⁵.

Powder X-ray diffraction. Powder X-ray diffraction measurements were carried out with a PANalytical Empyrean diffractometer using monochromated Cu/K α (λ = 1.5406 Å) at 45 kV and 40 mA.

Nuclear magnetic resonance. All the solid state nuclear magnetic resonance experiments were performed with magic angle spinning (MAS) on a Bruker Avance II 600 MHz wide-bore solid-state NMR spectrometer; Liquid NMR spectrums were recorded on a Bruker DPX 400 NMR spectrometer at 400 MHz for ¹H NMR. CDCl₃ was used as deuterated solvent. Tetramethylsilane (TMS) served as the internal standard (δ = 0.00 ppm) for ¹H NMR.

Raman spectra: All data were acquired using a Thermo Fisher DXR3xi Raman imaging microscope. Unless otherwise specified, Raman spectra were obtained with a 532 nm laser power of 0.3 mW. 2DCOF-A, -B and -AB were obtained with a 532 nm laser power of 2.0 mW.

Adsorption-desorption analysis. Gas adsorption isotherms (N₂/O₂/CO₂) for pressures in the range of 0-1.0 bar were obtained by a volumetric method using a Quantachrome autosorb-iQ2-MP gas adsorption analyzer, the gas adsorption isotherms of SO₂ collected by corrosive gas adsorption analyzer (BSD-PMC1). Breakthrough experiments data was collected by BSD-MAB (Multi-component Adsorption Breakthrough Curve Analyzer) used mass spectrometer as detector. Gas adsorption measurements were performed using ultra-high purity N₂, O₂, CO₂, SO₂.

Breakthrough Experiments. Breakthrough experiments was collected by BSD-MAB (Multi-component Adsorption Breakthrough Curve Analyzer) used mass spectrometer as detector. Transient breakthrough experiments for N₂/CO₂/O₂/SO₂ (81.8:15:3:0.2,

v/v/v) separation were performed in a fixed bed with a column (6 mm inner diameter × 150 mm) containing 1 g of the preactivated sample. Before filling in the column, the samples were activated at 373 K for 10 h under vacuum conditions. Afterward, the column was purged with a He flow for 4 h. The flow rates (40 mL min⁻¹ at 298 K, 1 bar) of gases were regulated by mass flow controllers. Then, the N₂/CO₂/O₂/SO₂ gas mixture with 40 mL min⁻¹ was introduced to the column packed with COF-AB-c, COF-A or COF-B. The sample was regenerated with a He flow (40 mL min⁻¹) at 373 K.

The capture capacity (also called capture amount) is determined as follows:⁶

$$q_i = \frac{v \times V_i\%}{22.4 \times m} \int_0^t \left(1 - \frac{C}{C_0}\right) dt$$

Where q_i is the capture amount of gas i (mmol g⁻¹), v is the flow rate of the gas mixture (mL min⁻¹), $V_i\%$ is molar fraction of gas i , t is the adsorption time (min), C_0 and C are the inlet and outlet gas concentration, respectively, and m is the mass of the adsorbent (g).

Theoretical Calculations

Adsorption Site Simulations. First-principle density functional theory (DFT) calculations were performed with the Forcite and Dmol³ modules of Materials Studio.⁷ The primitive cell of COF-AB4A, COF-AB-ABAB, COF-AB-1A3B, COF-AB-4B, COF-A and COF-B⁸ was used to reduce the size of the system for calculations. After transformed to *P1* space group, the C, N, O atoms were fixed. This initial structure was first optimized using Forcite module with universal force field (UFF). The convergence tolerances of energy, force and displacement were 2.0×10^{-5} kcal mol⁻¹, 1×10^{-3} kcal mol⁻¹ Å⁻¹, and 1×10^{-5} Å, respectively. Then Dmol³ module was used for DFT calculations. The perdue-Burke-Ernzerhof (PBE) functional was used under the generalized gradient approximation (GGA) functional with the double- ξ numerical polarization (DNP) basis set.⁹ The tolerances of energy, gradient, and displacement convergence were 1.0×10^{-5} hartree, 2×10^{-3} hartree Å⁻¹, and 5×10^{-3} Å, respectively. Thus, we obtained optimized COF-AB4A, COF-AB-ABAB,

COF-AB-1A3B, COF-AB-4B, COF-A and COF-B primitive cell. SO₂ molecule was directly optimized with Dmol³ using the previously described setting.¹⁰ These gas molecules were put into the COF-AB4A, COF-AB-ABAB, COF-AB-1A3B, COF-AB-4B, COF-A and COF-B primitive cell to find out the adsorption locations. The adsorption energies (ΔE) of gas molecules within COF-AB4A, COF-AB-ABAB, COF-AB-1A3B, COF-AB-4B, COF-A and COF-B were calculated by $\Delta E = E_{\text{gas}} + E_{\text{COF}} - E_{\text{gas-COF}}$, where E_{gas} , E_{COF} , and $E_{\text{gas-COF}}$ are the total energies of the gas molecules, the cages, and COF-gas systems at their optimized geometries, respectively.¹¹

Grand Canonical Monte Carlo Simulations. The grand canonical Monte Carlo (GCMC) simulations were performed with the Sorption module of Materials Studio.¹² The partial charges of SO₂ was calculated using the ESP method.¹⁰ The previously described Dmol³ optimized COF-A, COF-B and COF-AB-c primitive cell was used. The partial charges of COF-A, COF-B and COF-AB-c were calculated with the charge equilibration (Qeq) method.¹³ Atoms in COF-A, COF-B and COF-AB-c were fixed during GCMC simulations. The density distribution of the gases on the MOFs were calculated using the fixed pressure task mode of Sorption module under 298 K and 100 kPa (SO₂ fugacity: 100 kPa). The equilibration steps, production steps and temperature were 1×10^6 , 1×10^7 , and 298 K, respectively. The parameters of all-atom in the framework were from DREIDING force field.^{11,14} The van der Waals interactions with a cutoff 18.5 Å were depicted by the Lennard-Jones potential.

Procedure for ¹H NMR analysis of digested single-crystal COFs with different components

We screened deuterated reagents and finally selected deuterated chloroform because it can completely hydrolyze imines and well dissolve linkers. In general, 1 mg of dried COF was digested and dissolved with sonication in 0.5 mL of diluted DCl solution (prepared from 100 μL of 36% DCl/D₂O solution (Aldrich) and 5 mL DMSO-*d*₆ or CDCl₃). The digested solution was used directly for ¹H-NMR.

Section 1. Synthesis and characterization of multicomponent COFs

Supplementary Table 1. Summary of the synthesized isorecticular COFs and aCOFs.

COFs	Aldehyde monomers (0.4 mmol)	Yield/mg
COF-A	1, 4-phthalaldehyde (A , 53.65 mg)	104 (90%)
COF-B	2,5-dimethoxyterephthalaldehyde (B , 66.45 mg)	118 (92%)
COF-C	2,5-bis(prop-2-yn-1-yloxy)terephthalaldehyde (C , 96.89 mg)	136 (86%)
COF-D	2,3,5,6-tetrafluoroterephthalaldehyde (D , 82.44 mg)	134 (93%)
COF-E	thieno[3,2-b]thiophene-2,5-dicarboxaldehyde (E , 78.50 mg)	105 (75%)
aCOF-F	2,5-dichloroterephthalaldehyde (F , 81.21 mg)	113 (79%)
aCOF-G	2,5-dibromoterephthalaldehyde (G , 116.77 mg)	139 (78%)
aCOF-H	2-nitroterephthalaldehyde (H , 71.61 mg)	108 (81%)

COF-ABs and COF-BAs

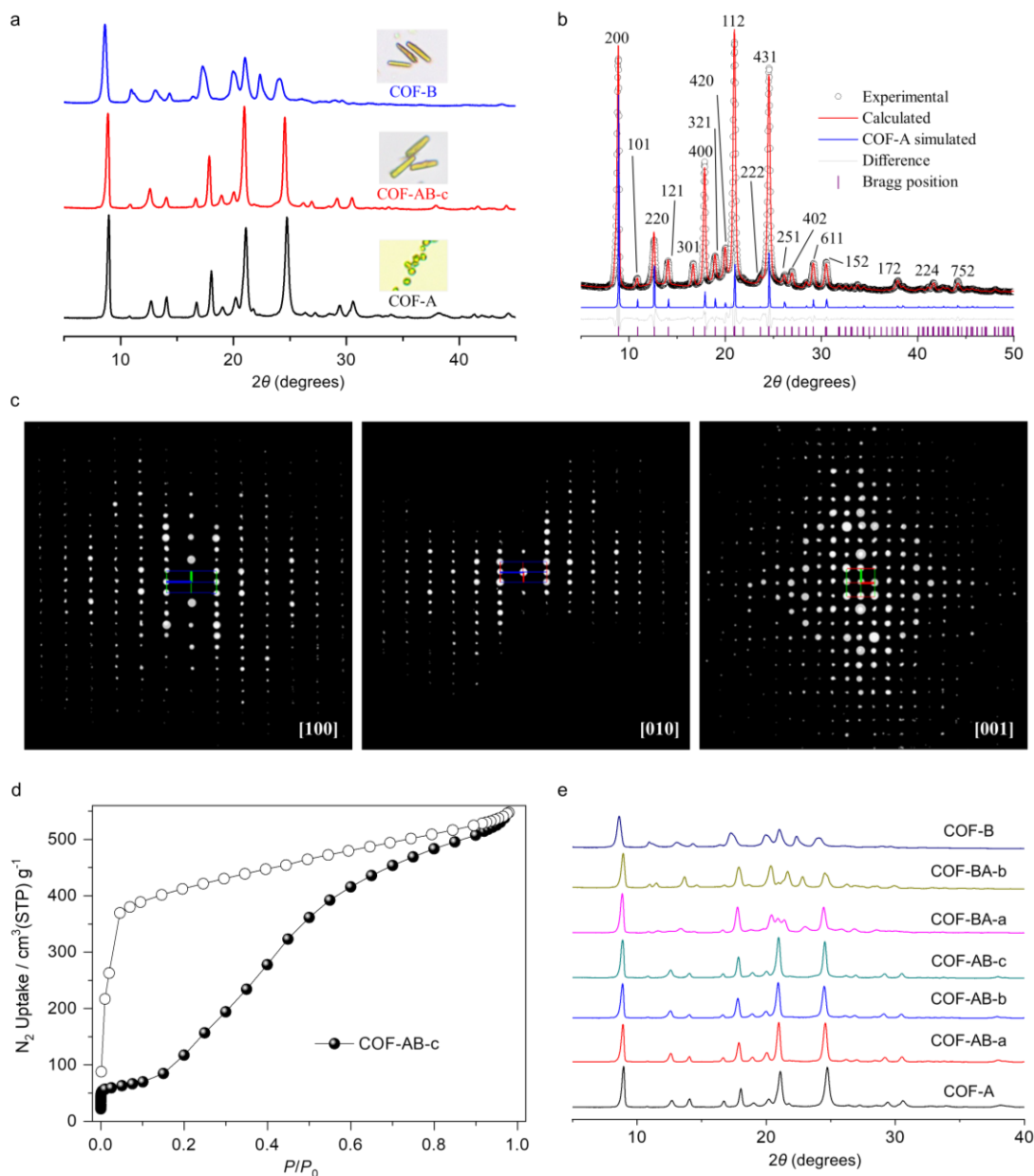
According to *General procedure A* terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg) and 2,5-dihydroxyterephthalaldehyde (**B**, 0.2 mmol, 33.2 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. The details of the products are summarized in **Supplementary Table 2**; *¹H NMR of the digested crystals* (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **B** δ 10.31 (s, 2H), 7.32 (s, 2H). Gram-scale production of COF-AB-c: C₁₆-GlyA (5.015 g, 16 mmol) and mixed linkers (Linker-A, 0.537 g, 4 mmol; Linker-B, 0.665 g, 4 mmol) was explored to gain highly crystalline COF-AB-c powder (Yield: 95%, 2.35 g). Recovery rate of C₁₆-GlyA: 94%, 4.71 g.

Supplementary Table 2. Summary of synthesis of COF-ABs and COF-BAs, their added stoichiometric link ratio, the ratio found in their crystals and yield.

COFs	A/mmol	B/mmol	Stoichiometric ratio (A:B)	Ratio in Crystal	Yield/mg
AB-a	0.32	0.08	1:0.25	1:0.27	90% (108)
AB-b	0.24	0.16	1:0.67	1:0.70	92% (112)
AB-c	0.20	0.20	1:1.00	1:0.83	92% (114)
BA-a	0.16	0.24	1:1.50	1:1.41	92% (115)
BA-b	0.08	0.32	1:4.00	1:2.31	95% (121)

Supplementary Table 3. Summary of Le Bail refinement parameters for COFs.

Entry	COFs	Space group	Label refined unit cell		<i>R</i> factors		Volume (Å ³)
			<i>a</i> = <i>b</i> (Å)	<i>c</i> (Å)	<i>R</i> _p (%)	<i>wR</i> _p (%)	
1	AB-a	<i>I</i> 4 ₁ / <i>a</i>	19.79392	8.93461	5.144	7.045	3501
2	AB-b	<i>I</i> 4 ₁ / <i>a</i>	19.89861	8.90570	6.589	9.128	3526
3	AB-c	<i>I</i> 4 ₁ / <i>a</i>	20.04649	8.91513	6.876	9.174	3583
4	BA-a	<i>C</i> 2/ <i>c</i>	<i>a</i> = 20.15523, <i>b</i> = 8.87182, <i>c</i> = 20.19753, <i>α</i> = <i>γ</i> = 90°, <i>β</i> = 82.06556		4.018	5.259	3577
5	BA-b	<i>C</i> 2/ <i>c</i>	<i>a</i> = 20.36805, <i>b</i> = 8.77625, <i>c</i> = 20.14672, <i>α</i> = <i>γ</i> = 90°, <i>β</i> = 80.57697		3.333	4.537	3553



Supplementary Figure 1. Single crystal structure of COF-AB-c and its N_2 adsorption isotherm. **a**, Powder XRD pattern and optical image of COF-A, COF-AB-c and COF-B. **b**, Structural refinement of COF-AB-c from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with an interpenetration degree of seven (blue). **c**, View along the [100], [010] and [001] directions of COF-AB-c and 3D reciprocal lattice reconstructed from the continuous rotational electron diffraction (cRED) dataset. **d**, N_2 adsorption isotherms of COF-AB-c at 77 K. **e**, Powder XRD pattern of COF-A, COF-ABs, COF-BAs and COF-B.

COF-AC

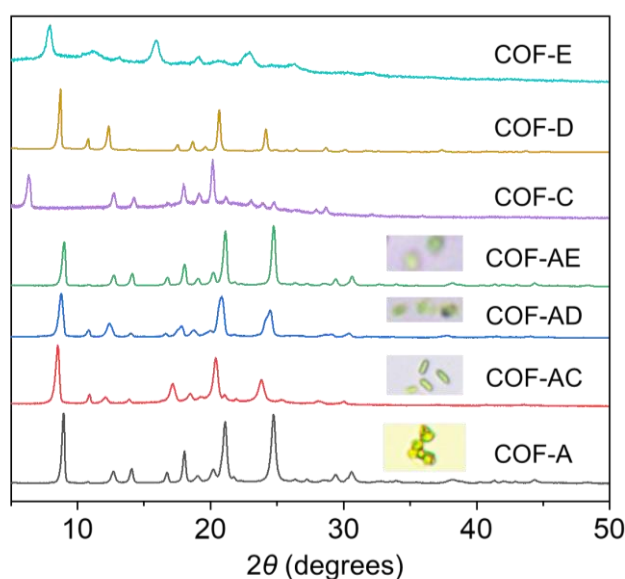
According to **General procedure A** terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg) and 2,5-bis(prop-2-yn-1-yloxy)terephthalaldehyde (**C**, 0.2 mmol, 48.4 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. Yield: 63% (87 mg). **¹H NMR of the digested crystals** (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **C** δ 10.52 (s, 2H), 7.62 (s, 2H), 4.88 (s, 4H), 2.59 (s, 2H). Molar ratio: A: C =1: 0.62.

COF-AD

According to **General procedure A** terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg) and 2,3,5,6-tetrafluoroterephthalaldehyde (**D**, 0.2 mmol, 41.2 mg) was added, reaction at 50 °C for 7 days to afford yellow crystal. Yield: 80% (103 mg). **¹H NMR of the digested crystals** (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **G** δ 10.39 (s, 2H). Molar ratio: A: D =1: 1.07.

COF-AE

According to **General procedure A** terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg) and thieno[3,2-b]thiophene-2,5-dicarbaldehyde (**E**, 0.2 mmol, 39.2 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. Yield: 50% (65 mg). **¹H NMR of the digested crystals** (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **E** δ 10.05 (s, 2H), 8.01 (s, 2H). Molar ratio: A: E=1: 0.02.



Supplementary Figure 2. Powder XRD pattern and optical image of COF-A, COF-AC, COF-AD, COF-AE, COF-C, COF-D, COF-E.

COF-AF

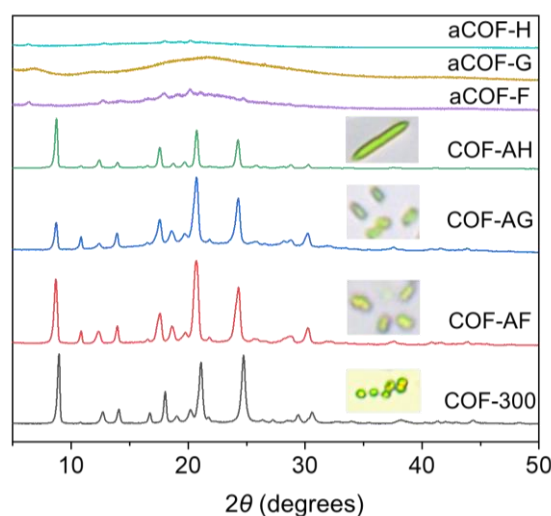
According to **General procedure A** terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg) and 2,5-dichloroterephthalaldehyde (**F**, 0.2 mmol, 40.6 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. Yield: 73% (96 mg). **¹H NMR of the digested crystals** (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **F** δ 10.49 (s, 2H), 8.03 (s, 2H). Molar ratio: A: F =1: 1.05.

COF-AG

According to **General procedure A** terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg) and 2,5-dibromoterephthalaldehyde (**G**, 0.2 mmol, 58.4 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. Yield: 53% (79 mg). **¹H NMR of the digested crystals** (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **G** δ 10.37 (s, 2H), 8.18 (s, 2H). Molar ratio: A: G=1: 0.6.

COF-AH

According to **General procedure A** terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg) and 2-nitroterephthalaldehyde (**H**, 0.2 mmol, 35.8 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. Yield: 67% (84 mg). **¹H NMR of the digested crystals** (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **H** δ 10.52 (s, 1H), 10.20 (s, 1H), 8.65 (s, 1H), 8.32 (d, *J* = 7.8 Hz, 1H), 8.14 (d, *J* = 7.8 Hz, 1H). Molar ratio: A: H =1: 1.08.



Supplementary Figure 3. Powder XRD pattern and optical image of COF-A, COF-AF, COF-AG, COF-AH, aCOF-F, aCOF-G, aCOF-H.

Supplementary Table 4. Summary of Le Bail refinement parameters for COFs.

Entry	COFs	Space group	Label refined unit cell		<i>R</i> factors		Volume (Å ³)
			<i>a</i> = <i>b</i> (Å)	<i>c</i> (Å)	<i>R</i> _p (%)	<i>wR</i> _p (%)	
1	AC	<i>I</i> 4 ₁ / <i>a</i>	20.61348	8.78601	4.788	6.287	3501
2	AD	<i>I</i> 4 ₁ / <i>a</i>	20.04649	8.91513	6.876	9.174	3526
3	AE	<i>I</i> 4 ₁ / <i>a</i>	19.69815	8.94200	3.929	5.359	3583
4	AF	<i>I</i> 4 ₁ / <i>a</i>	20.15012	8.89004	7.675	9.707	3609
5	AG	<i>I</i> 4 ₁ / <i>a</i>	20.03433	8.90789	5.943	8.141	3575
6	AH	<i>I</i> 4 ₁ / <i>a</i>	20.13310	8.87963	5.457	7.121	3733

COF-ABDIs

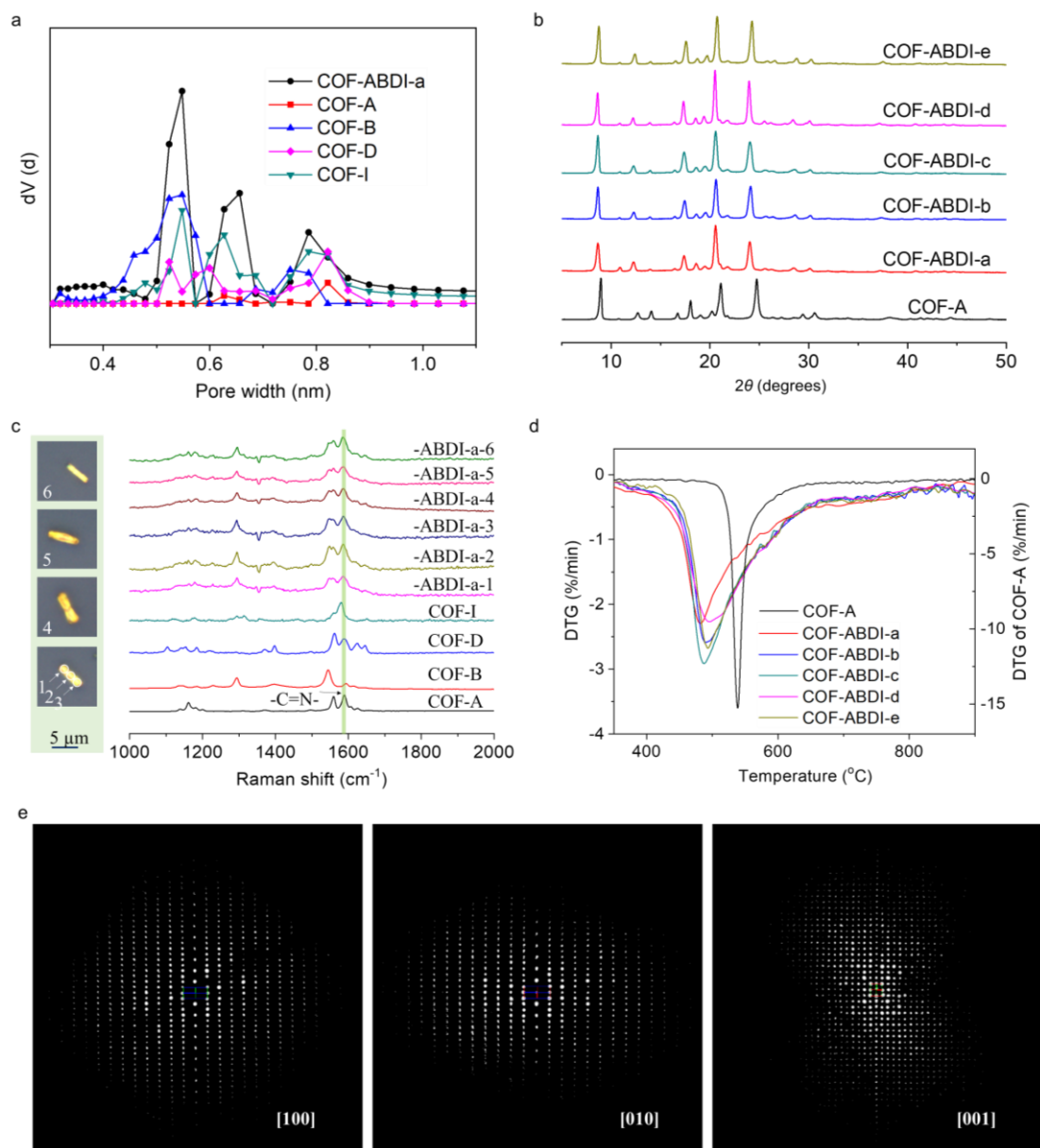
According to *General procedure A* terephthalaldehyde (**A**, 0.1 mmol, 13.4 mg), 2,5-dihydroxyterephthalaldehyde (**B**, 0.1 mmol, 16.6 mg), 2,3,5,6-tetrafluoroterephthalaldehyde (**D**, 0.1 mmol, 20.6 mg) and 2,5-dimethoxyterephthalaldehyde (**I**, 0.1 mmol, 19.4 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. The details of the products are summarized in **Supplementary Table 3; ¹H NMR of the digested crystals** (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **B** δ 10.31 (s, 2H), 7.32 (s, 2H). **D** δ 10.39 (s, 2H). **I** δ 10.53 (s, 2H), 7.48 (s, 2H), 3.97 (s, 6H). Gram-scale production of COF-ABDI-a: C₁₆-GlyA (5.015 g, 16 mmol) and mixed linkers (Linker-A: 0.268 g, 2 mmol; Linker-B: 0.332 g, 2 mmol; Linker-D: 0.412 g, 2 mmol; Linker-I: 0.388 g, 2 mmol) were explored to gain highly crystalline powder (Yield: 91%, 2.32 g). Recovery rate of C₁₆-GlyA: 94%, 4.70 g.

Supplementary Table 5. Summary of synthesis of COF-ABDIs, their added stoichiometric link ratio, the ratio found in their crystals and yield.

COFs	A/ mmol	B/ mmol	I/ mmol	D/ mmol	Stoichiometric ratio (A:B:I:D)	Ratio in Crystal	Yield/mg
ABDI-a	0.10	0.10	0.10	0.10	1:1:1:1	1:1.09:0.34:1.04	77% (103)
ABDI-b	0.20	0.067	0.067	0.067	1:0.34:0.34:0.34	1:0.36:0.17:0.35	86% (110)
ABDI-c	0.20	0.1005	0.067	0.0335	1:0.51:0.34:0.17	1:0.55:0.19:0.16	91% (115)
ABDI-d	0.20	0.0335	0.1005	0.067	1:0.17:0.51:0.34	1:0.17:0.30:0.34	87% (112)
ABDI-e	0.20	0.067	0.0335	0.1005	1:0.34:0.17:0.51	1:0.39:0.07:0.56	92% (119)

Supplementary Table 6. Summary of Le Bail refinement parameters for COFs.

Entry	COFs	Space group	Label refined unit cell		<i>R</i> factors		Volume (Å ³)
			<i>a</i> = <i>b</i> (Å)	<i>c</i> (Å)	<i>R</i> _p (%)	<i>wR</i> _p (%)	
1	ABDI-a	<i>I</i> 4 ₁ / <i>a</i>	20.40096	8.85984	3.798	5.217	3687
2	ABDI-b	<i>I</i> 4 ₁ / <i>a</i>	20.22342	8.84221	5.997	8.416	3616
3	ABDI-c	<i>I</i> 4 ₁ / <i>a</i>	20.29508	8.83106	5.415	7.657	3637
4	ABDI-d	<i>I</i> 4 ₁ / <i>a</i>	20.42879	8.85113	3.744	5.394	3694
5	ABDI-e	<i>I</i> 4 ₁ / <i>a</i>	20.16604	8.88698	5.494	7.383	3614
6	ABCDEFGHI	<i>I</i> 4 ₁ / <i>a</i>	19.98068	8.93775	6.134	8.160	3568



Supplementary Figure 4. Single crystal COF-ABDIs. **a**, Pore size distribution calculated from CO₂ adsorption isotherm at 273 K. **b**, Powder XRD pattern of COF-A and COF-ABDIs (-a, -b, -c, -d and -e represent different ratios of the four linkers (A: B: E: I)) with the specific ratios summarized in Extended Data Table 2. **c**, Optical image and Raman spectra of marked points of the COF-ABDI-a crystals. **d**, Differential thermal gravities of COF-A and COF-ABDIs. **e**, View along the [100], [010] and [001] directions of COF-ABDI-a and 3D reciprocal lattice reconstructed from cRED dataset.

COF-ABCDEFGHI

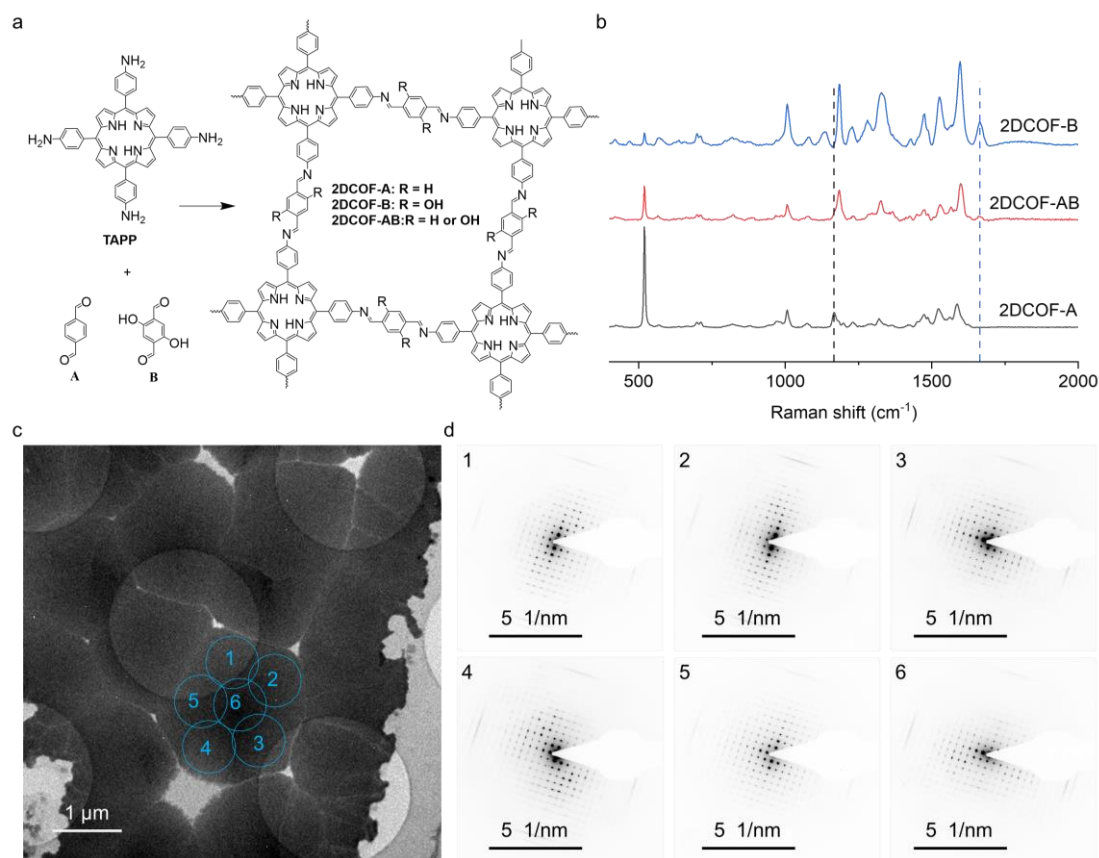
According to *General procedure A* terephthalaldehyde (**A**, 0.2 mmol, 26.8 mg), 2,5-dihydroxyterephthalaldehyde (**B**, 0.05 mmol, 4.2 mg), 2,5-bis(prop-2-yn-1-yloxy)terephthalaldehyde (**C**, 0.05 mmol, 6.1 mg), 2,3,5,6-tetrafluoroterephthalaldehyde (**D**, 0.05 mmol, 5.2 mg), thieno[3,2-*b*]thiophene-2,5-dicarbaldehyde (**E**, 0.05 mmol, 4.9), 2,5-dichloroterephthalaldehyde (**F**, 0.05 mmol, 5.1 mg), 2,5-dibromoterephthalaldehyde (**G**, 0.05 mmol, 7.3 mg), 2-nitroterephthalaldehyde (**H**, 0.05 mmol, 4.5 mg) and 2,5-dimethoxyterephthalaldehyde (**I**, 0.05 mmol, 4.9 mg) were added, reaction at 50 °C for 7 days to afford yellow crystal. Yield: 85% (113 mg). *¹H NMR of the digested crystals* (400 MHz, CDCl₃). **A** δ 10.15 (s, 2H), 8.13 (s, 4H). **B** δ 10.31 (s, 2H), 7.32 (s, 2H). **C** δ 10.52 (s, 2H), 7.62 (s, 2H), 4.88 (s, 4H), 2.59 (s, 2H). **D** δ 10.39 (s, 2H). **E** δ 10.05 (s, 2H), 8.01 (s, 2H). **F** δ 10.49 (s, 2H), 8.03 (s, 2H). **G** δ 10.37 (s, 2H), 8.18 (s, 2H). **H** δ 10.52 (s, 1H), 10.20 (s, 1H), 8.65 (s, 1H), 8.32 (d, *J* = 7.8 Hz, 1H), 8.14 (d, *J* = 7.8 Hz, 1H). **I** δ 10.53 (s, 2H), 7.48 (s, 2H), 3.97 (s, 6H). Molar ratio: A: B: C: D: E: F: G: H: I = 1: 0.12: 0.04: 0.15: 0.03: 0.08: 0.05: 0.11: 0.05. Gram-scale production of COF-ABCDEFGHI: C₁₆-GlyA (5.015 g, 16 mmol) and mixed linkers (Linker-A: 0.537 g, 4 mmol; Linker-B: 0.083 g, 0.5 mmol; Linker-C: 0.121 g, 0.5 mmol; Linker-D: 0.103 g, 0.5 mmol; Linker-E: 0.098 g, 0.5 mmol; Linker-F: 0.102 g, 0.5 mmol; Linker-G: 0.146 g, 0.5 mmol; Linker-H: 0.090 g, 0.5 mmol; Linker-I: 0.0097 g, 0.5 mmol) was explored to gain highly crystalline powder (Yield: 91%, 2.39 g). Recovery rate of C₁₆-GlyA: 94%, 4.73 g.

Synthesis procedures of 2DCOF-AB

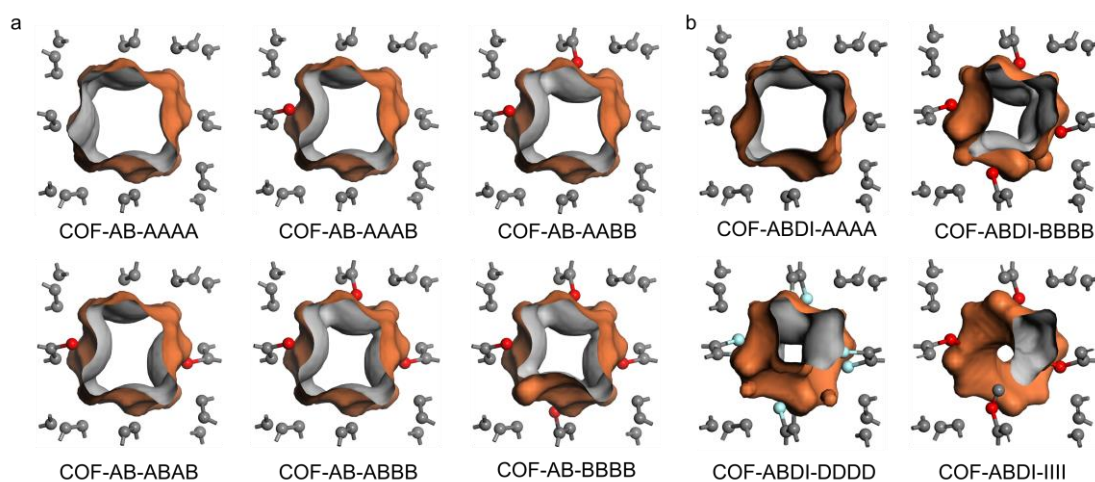
PTSA (*p*-toluenesulfonic acid) was dissolved in water to prepare 2 mol l⁻¹ PTSA aqueous solutions, then 5,10,15,20-tetrakis (4-aminophenyl)-21H,23H-porphyrin (TAPP) was dissolved in 2 mol l⁻¹ PTSA solution to prepare 1 mg ml⁻¹ TAPP solution. Terephthalaldehyde and 2,5-dihydroxyterephthalaldehyde were dissolved in ethylene glycol solution to prepare 1 mg ml⁻¹ terephthalaldehyde solution and 1 mg

ml⁻¹ 2,5-dihydroxyterephthalaldehyde solution, respectively. All solutions were sonicated for 30 minutes before use.

In a general synthetic procedure, 25 ml of water was added to a Petri dish (diameter of 56 mm and height of 16 mm). An ethanol solution of (*S*)-3,3-dimethyl-2-palmitamidobutanoic acid (C₁₆-*L*-*t*LeuA, 60 μ l, 1 mg ml⁻¹) was spread at the air–water interface and left undisturbed for 30 min. A solution of TAPP protonated with 2 mol l⁻¹ PTSA (200 μ l, 0.3 μ mol) was added to the Petri dish and left undisturbed for 60 min. Then 200 μ l of terephthalaldehyde solution and 200 μ l of 2,5-dihydroxyterephthalaldehyde solution were slowly injected into the vessel. Single-crystal 2DCOF-AB was obtained after one week. The films were then transferred onto substrates of interest from the air–water surface and cleaned with water and chloroform before characterization.



Supplementary Figure 5. Bicomponent single-crystal 2DCOF-AB. **a**, Chemical structure of knot TAPP (5,10,15,20-Tetrakis(4-aminophenyl)-21H,23H-porphyrin), linker-A (terephthalaldehyde), linker-B (2,5-dihydroxyterephthalaldehyde), 2DCOF-A, 2DCOF-B and 2DCOF-AB. **b**, Raman spectra of film of 2DCOF-A, 2DCOF-B, and 2DCOF-AB on silicon nitride substrate. **c**, Transmission electron microscopy image of film of 2DCOF-AB suspended over copper grids. **d**, Selected area electron diffraction of marked area with numbers in (c).

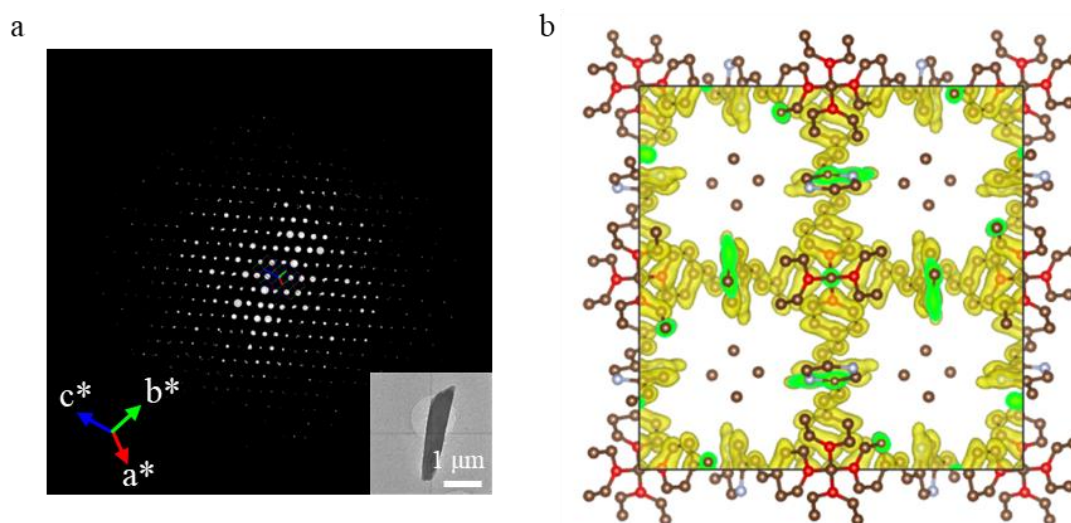


Supplementary Figure 6. Pore structure of COF-ABs and COF-ABDIs. a, b, Accessible Connolly surface of several pore structures of COF-AB (a) and COF-ABDI (b) determined by using a probe with a radius of 1.0 Å. The last part of the name of the crystals represents the linker and their sequence at the pore wall of the lattice.

Supplementary Table 7. Summary of pore size and unit cell free volume of COF-ABs and COF-ABDIs calculated from single crystal structure.

COFs	Pore Size/ Å	Free volume/ Å ³	COFs	Pore Size/ Å	Free volume/ Å ³
COF-AB-AAAA	5.89×5.89	898.64	COF-ABDI-AAAA	5.70×5.70	733.45
COF-AB-AAAB	5.39×5.89	870.43	COF-ABDI -BBBB	5.19×5.19	707.28
COF-AB-AABB	5.39×5.39	857.89	COF-ABDI -DDDD	5.30×5.30	552.18
COF-AB-ABAB	5.39×5.89	859.85	COF-ABDI -III	4.05×4.05	217.62
COF-AB-ABBB	5.39×5.39	813.60			
COF-AB-BBBB	5.39×5.39	771.67			

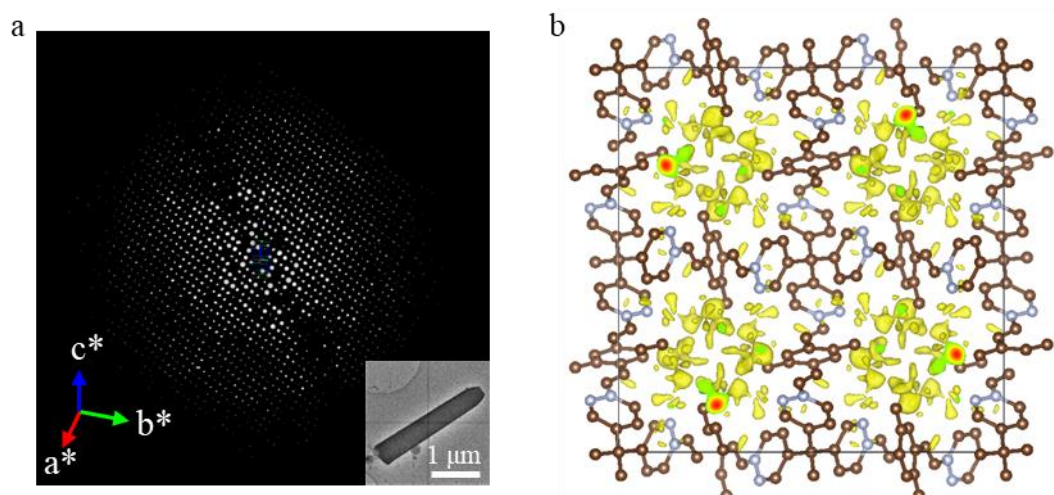
Section 2. Continuous rotation electron diffraction (cRED) analysis



Supplementary Figure 7. **a**, The 3D reciprocal lattice of COF-AB-c reconstructed from the cRED data. Overview of 3D reciprocal lattice. Inset: TEM image of the crystal ($2.5 \times 0.5 \times 0.5 \mu\text{m}^3$), from which the cRED data was collected. **b**, The initial structure of COF-AB-c obtained from structure solution with cRED data.

Supplementary Table 8. Crystallographic data and structure refinement details of COF-AB-c.

Name	COF-AB-c
Molecular formula	C ₄₁ N ₄ O ₄ H ₂₈
Data collection method	cRED
Temperature / K	100
Wavelength / Å	0.0251
Tilt range / °	-47.75 ~ 47.55
Crystal system	Tetragonal
Space group	<i>I</i> 4 ₁ / <i>a</i>
Unit cell dimensions	$a = b = 19.92 \text{ Å}, c = 8.64 \text{ Å},$ $\alpha = \beta = \gamma = 90^\circ$
Cell volume / Å ³	3428.407
<i>Z</i>	4
Resolution / Å	0.92
<i>R</i> _{int} / %	13.98
Completeness / %	88.90
Total/Unique number of reflections	3357/1133
<i>R</i> / %	32.38
<i>wR</i> / %	42.65



Supplementary Figure 8. **a**, The 3D reciprocal lattice of COF-ABDI-a reconstructed from the first cRED dataset. Overview of 3D reciprocal lattice. Inset: TEM image of the crystal ($3.3 \times 0.5 \times 0.5 \mu\text{m}^3$), from which the cRED data was collected. **b**, The initial structure of COF-ABDI-a obtained from structure solution with cRED data.

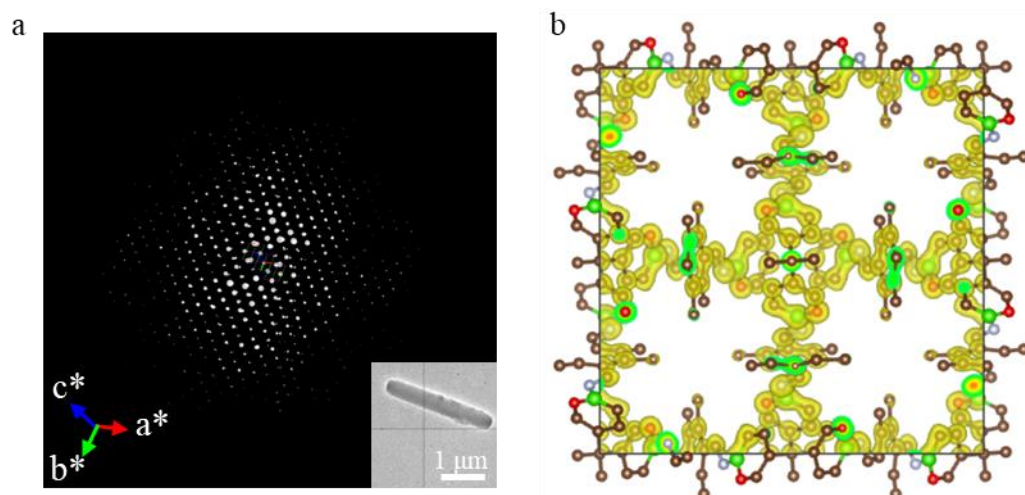
Three cRED datasets of COF-ABDI-a were collected from different crystals. The data processing results are given in **Supplementary Table 9**. Structure solution and refinement were conducted on the merged one dataset. Crystallographic data and structure refinement details of COF-ABDI-a are given in **Supplementary Table 10**.

Table 9. Data processing results of two datasets and the merged one.

Dataset	Rint / %	Resolution/ Å	Completeness / %	Mean (I/sigma)	No. Total Refls	Cell Dimension / Å	Volume	Tilt Range / °
1	15.49	0.75	93.1	3.53	6257	19.43, 8.69	3280.69	103.68
2	14.00	0.70	85.4	4.19	7174	19.27, 8.74	3245.45	111.28
Merged	19.31	0.75	99.8	5.03	12476	19.34, 8.74	3269.07	—

Supplementary Table 10. Crystallographic data and structure refinement details performed on the merged cRED dataset of COF-ABDI-a.

Name	COF-ABDI-a
Molecular formula	$\text{C}_{41.39}\text{N}_4\text{O}_{1.65}\text{F}_{2.4}\text{H}_{26.38}$
Data collection method	cRED
Temperature / K	100
Wavelength / Å	0.0251
Crystal system	Tetragonal
Space group	$I4_1/a$
Unit cell dimensions	$a = b = 19.34 \text{ Å}, c = 8.74 \text{ Å},$ $\alpha = \beta = \gamma = 90^\circ$
Cell volume / Å ³	3269.07
<i>Z</i>	4
Resolution / Å	0.75
R_{int}	19.31
Completeness / %	99.80
R / %	23.86
wR / %	29.17

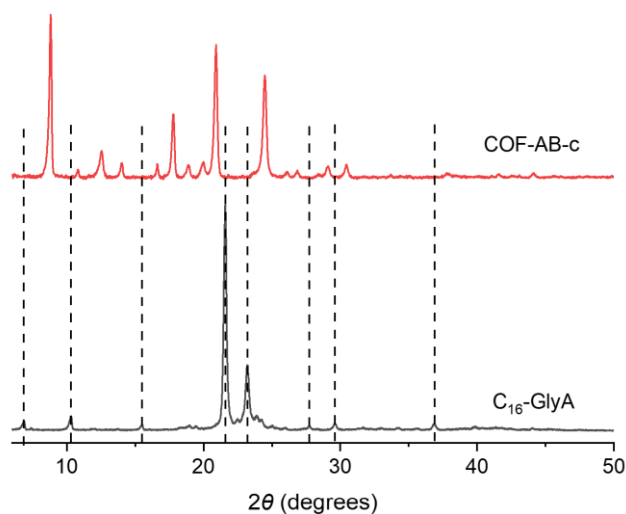


Supplementary Figure 9. **a**, The 3D reciprocal lattice of COF-ABCDEFGHI reconstructed from the cRED data. Overview of 3D reciprocal lattice. Inset: TEM image of the crystal ($2.1 \times 0.5 \times 0.5 \mu\text{m}^3$), from which the cRED data was collected. **b**, The initial structure of COF-ABCDEFGHI obtained from structure solution with cRED data.

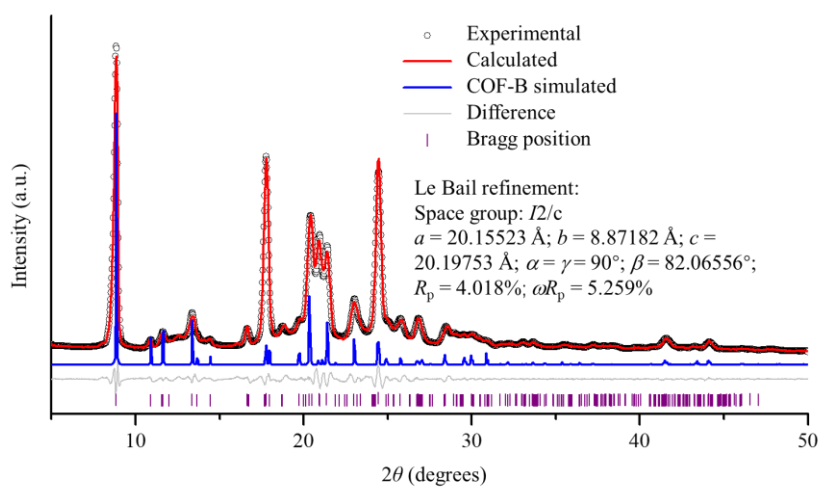
Supplementary Table 11. Crystallographic data and structure refinement details of COF-ABCDEFGHI.

Name	COF-ABCDEFGHI
Data collection method	cRED
Temperature / K	100
Wavelength / Å	0.0251
Tilt range / °	-48.13 ~ 55.26
Crystal system	Tetragonal
Space group	$I4_1/a$
Unit cell dimensions	$a = b = 19.95 \text{ Å}, c = 8.67 \text{ Å},$ $\alpha = \beta = \gamma = 90^\circ$
Cell volume / Å ³	3450.682
<i>Z</i>	4
Resolution / Å	0.80
R_{int} / %	9.66
Completeness / %	96.1
Total/Unique number of reflections	4987/1703
R / %	24.86
wR / %	32.01

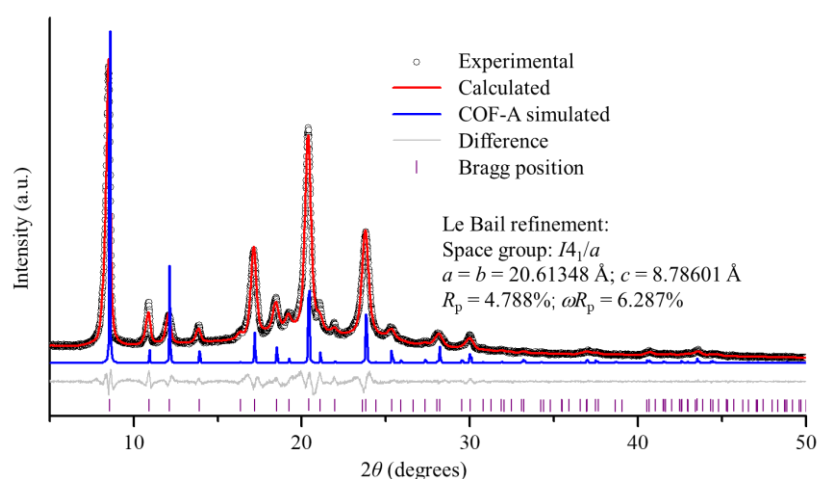
Section 3. Powder XRD analysis



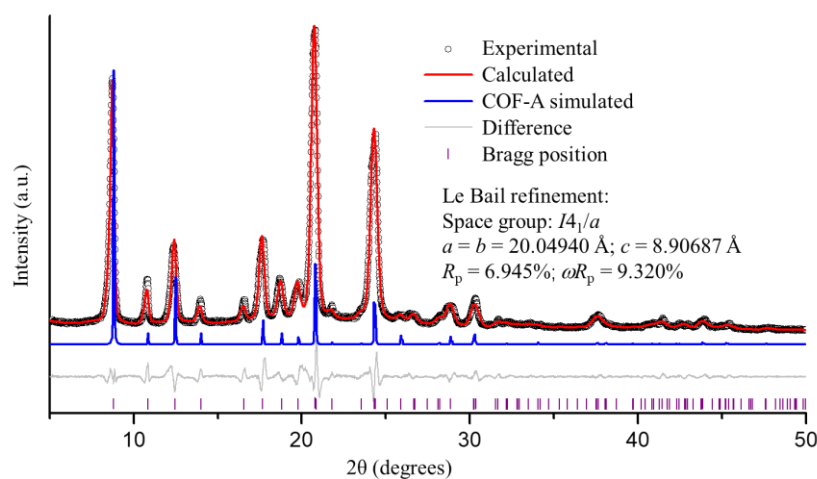
Supplementary Figure 10. Powder XRD pattern of C₁₆-GlyA and COF-AB-c.



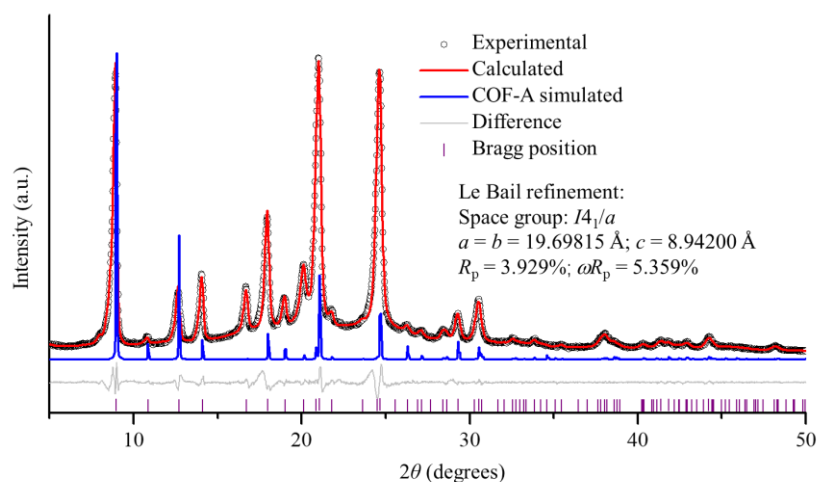
Supplementary Figure 11. Structural refinement of COF-BA-a from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).



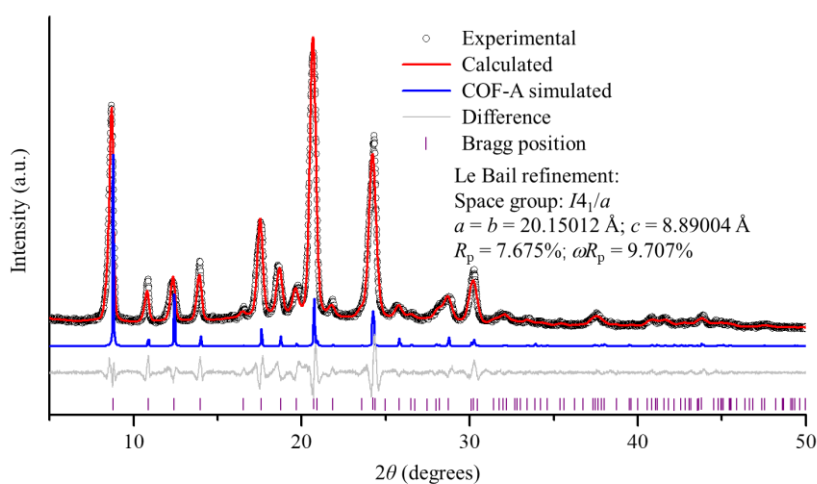
Supplementary Figure 12. Structural refinement of COF-AC from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).



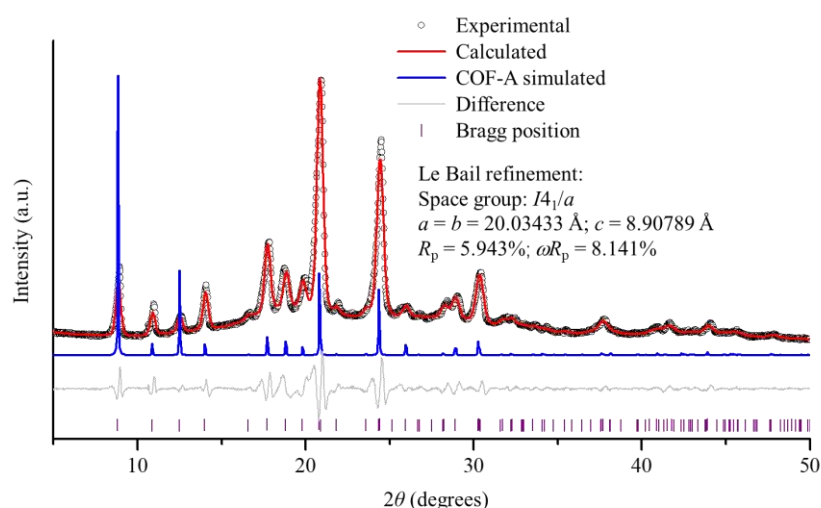
Supplementary Figure 13. Structural refinement of COF-AD from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).



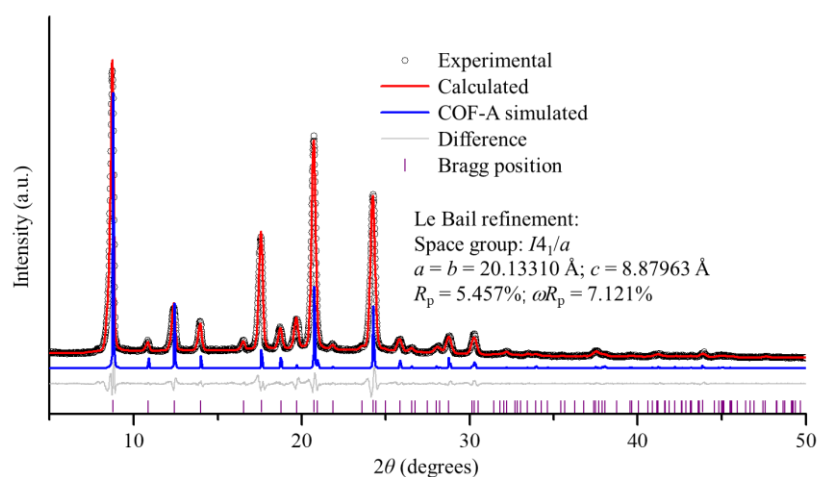
Supplementary Figure 14. Structural refinement of COF-AE from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).



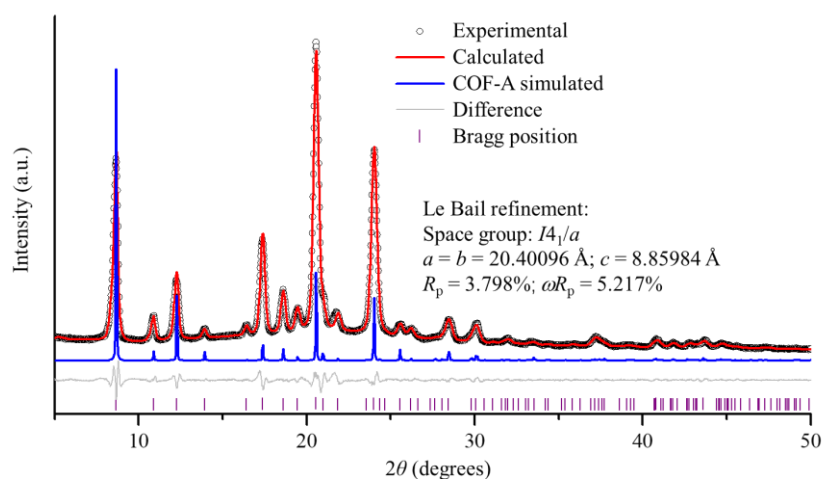
Supplementary Figure 15. Structural refinement of COF-AF from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).



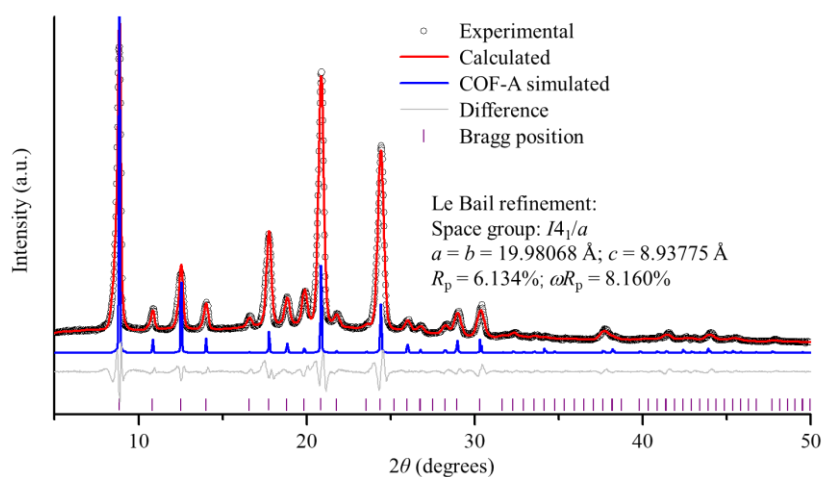
Supplementary Figure 16. Structural refinement of COF-AG from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).



Supplementary Figure 17. Structural refinement of COF-AH from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).

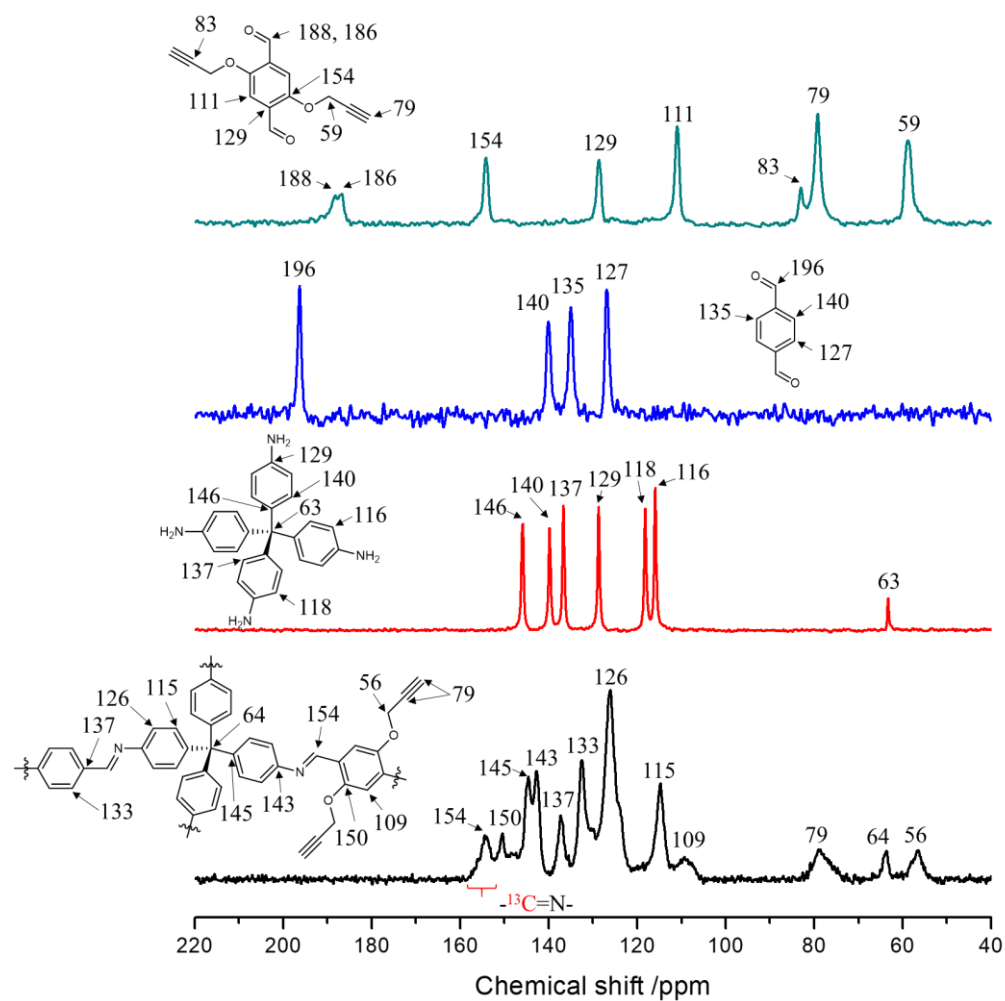


Supplementary Figure 18. Structural refinement of COF-ABDI-a from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).

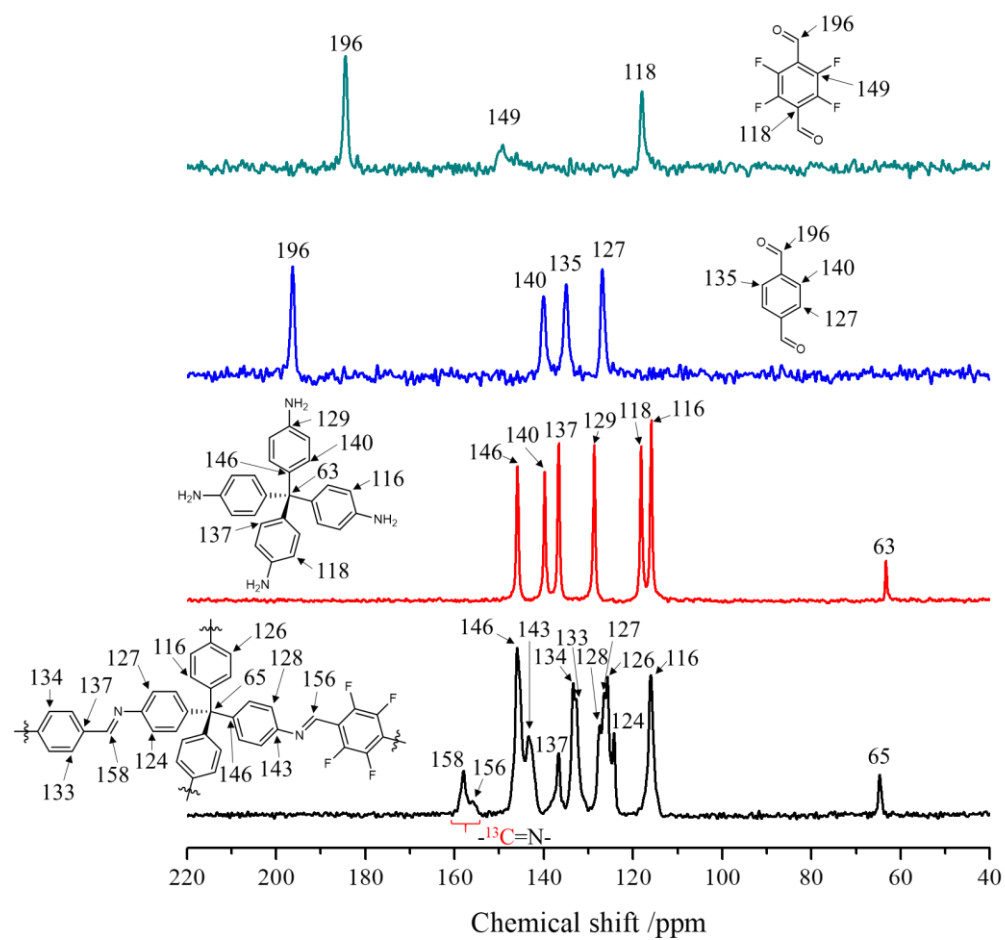


Supplementary Figure 19. Structural refinement of COF-ABCDEFGHI from powder XRD data analysis displaying the indexed experimental pattern (black), Le Bail refinement (red), and the simulated pattern of COF-A with interpenetration degree of seven (blue).

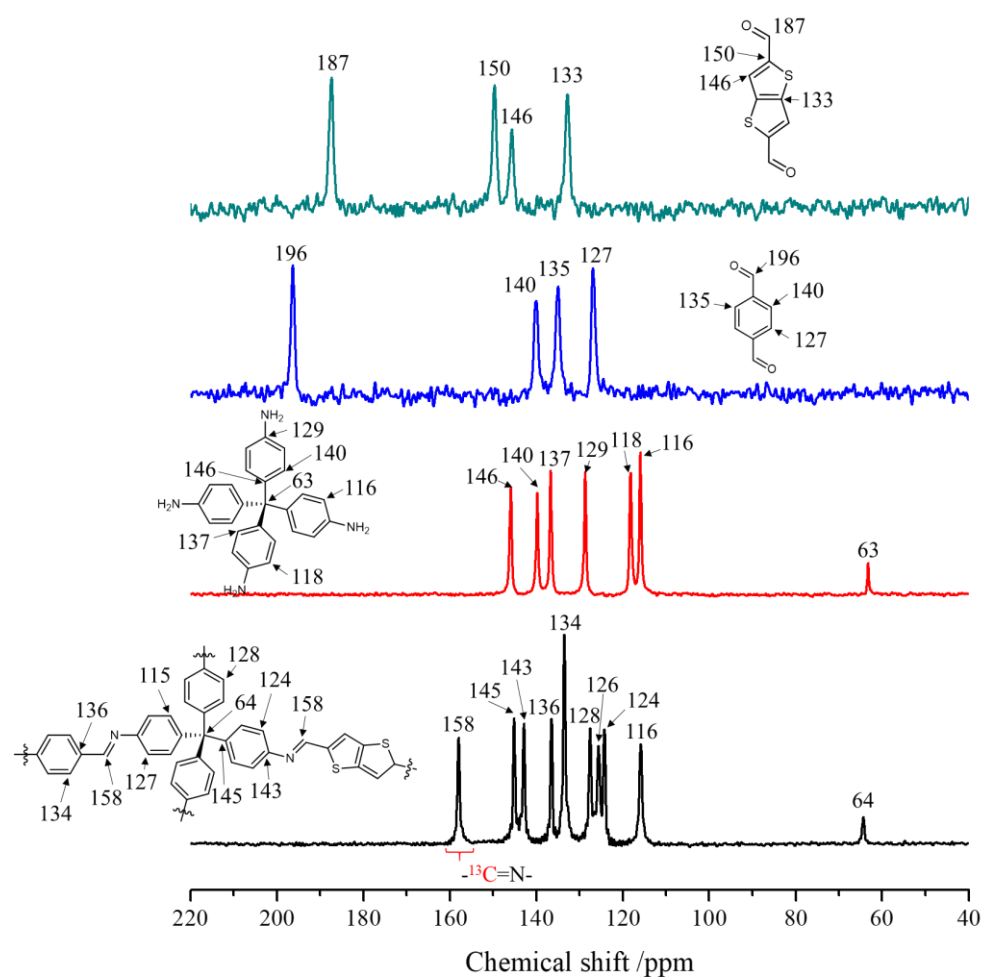
Section 4. Solid-state nuclear magnetic resonance spectroscopic analysis



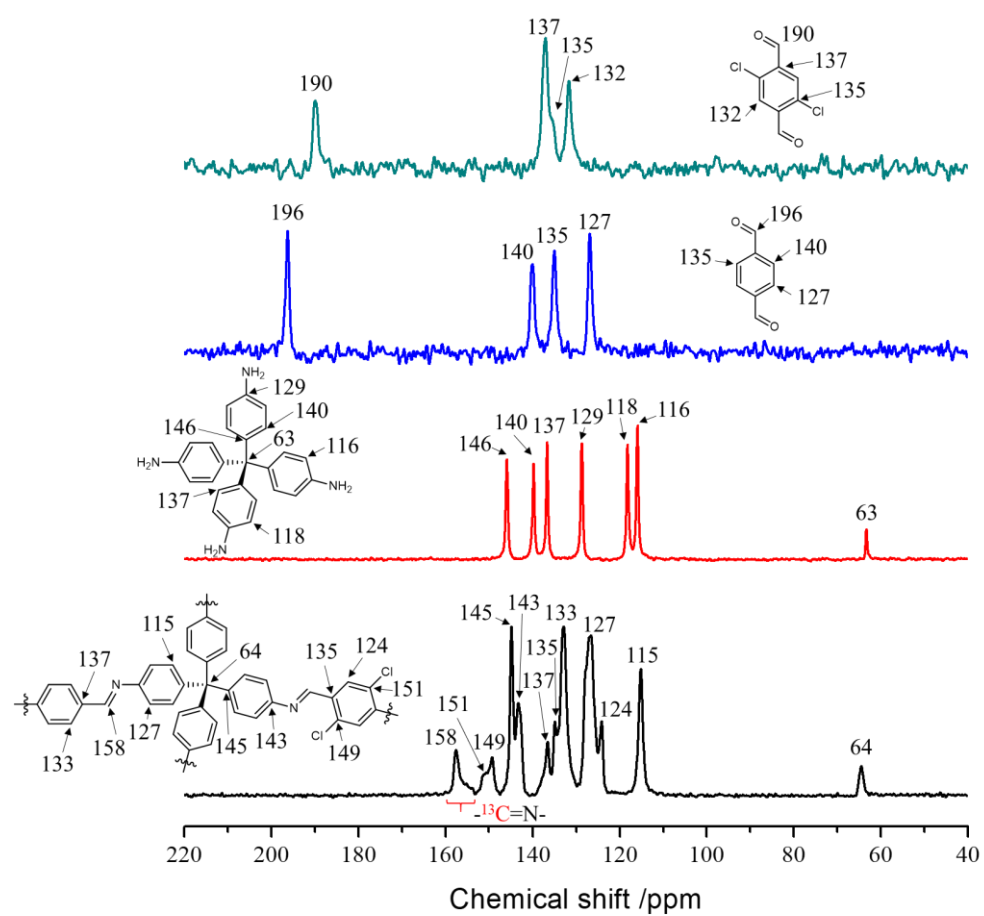
Supplementary Figure 20. ^{13}C CP/MAS spectra of COF-AC, black; knot (TAM), red; linker-A, blue; linker-C, dark cyan.



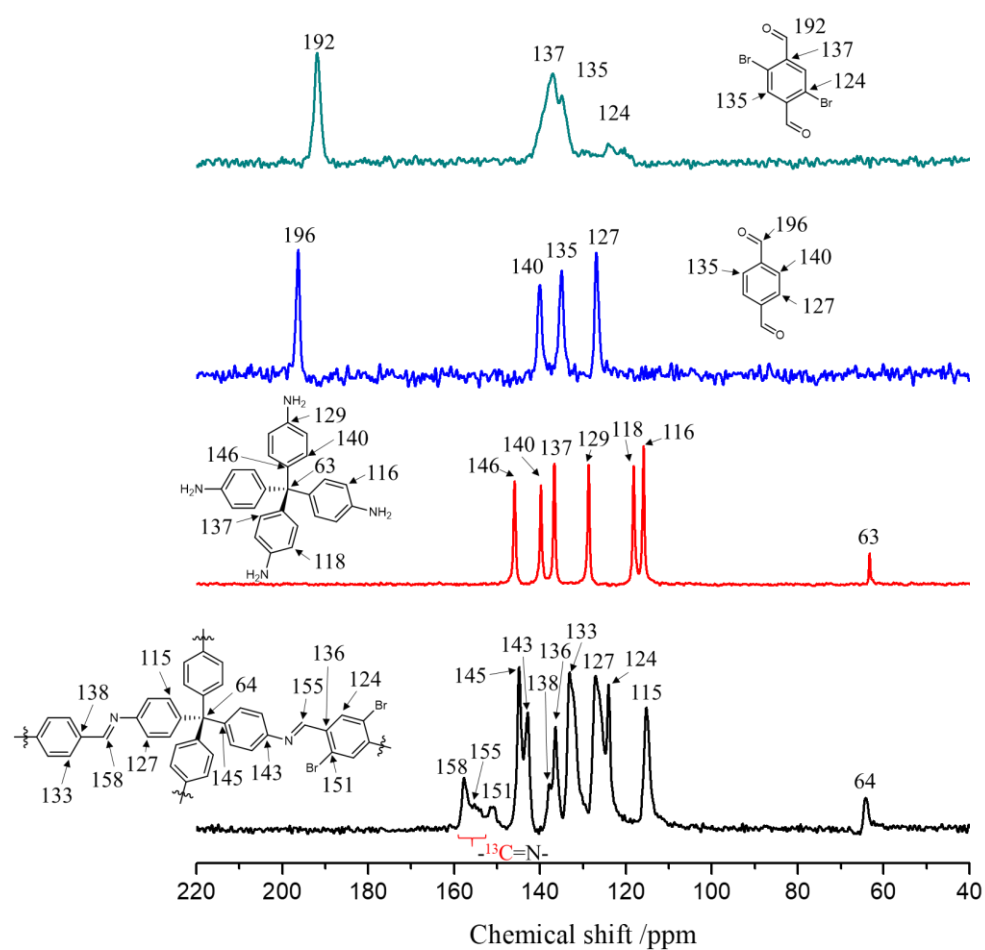
Supplementary Figure 21. ^{13}C CP/MAS spectra of COF-AD, black; knot (TAM), red; linker-A, blue; linker-D, dark cyan.



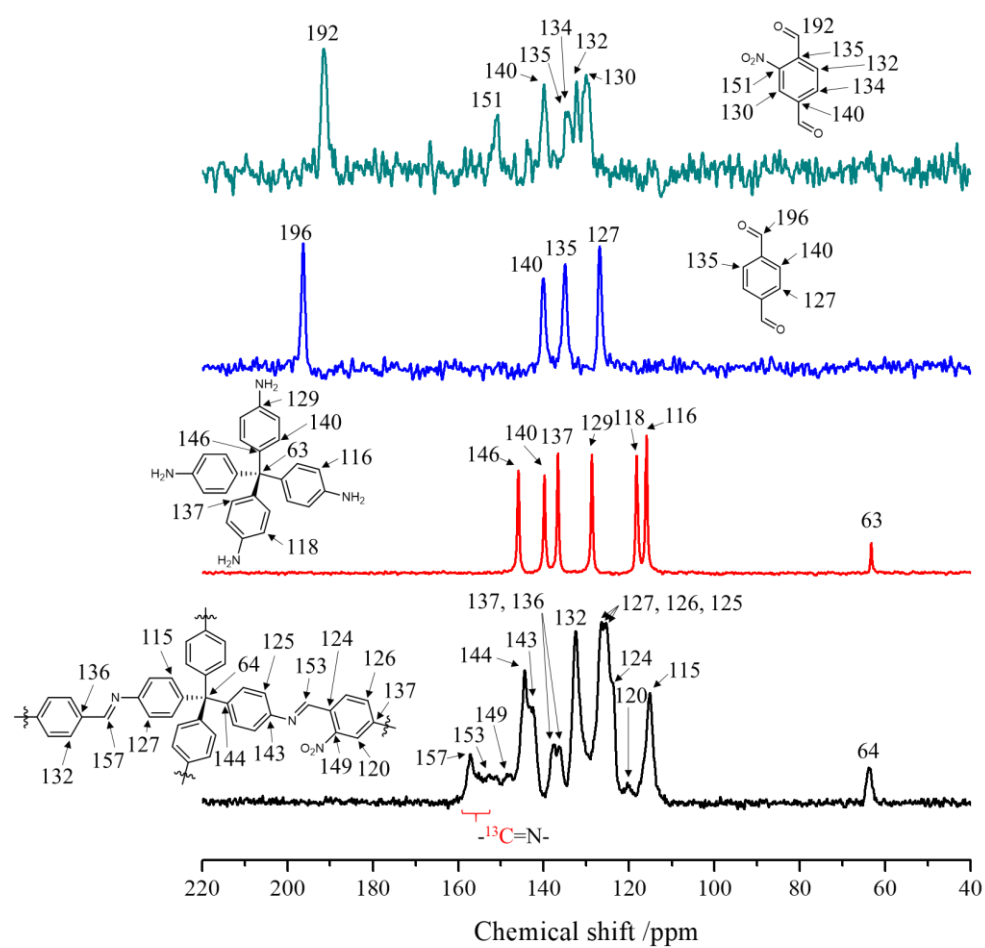
Supplementary Figure 22. ^{13}C CP/MAS spectra of COF-AE, black; knot (TAM), red; linker-A, blue; linker-E, dark cyan.



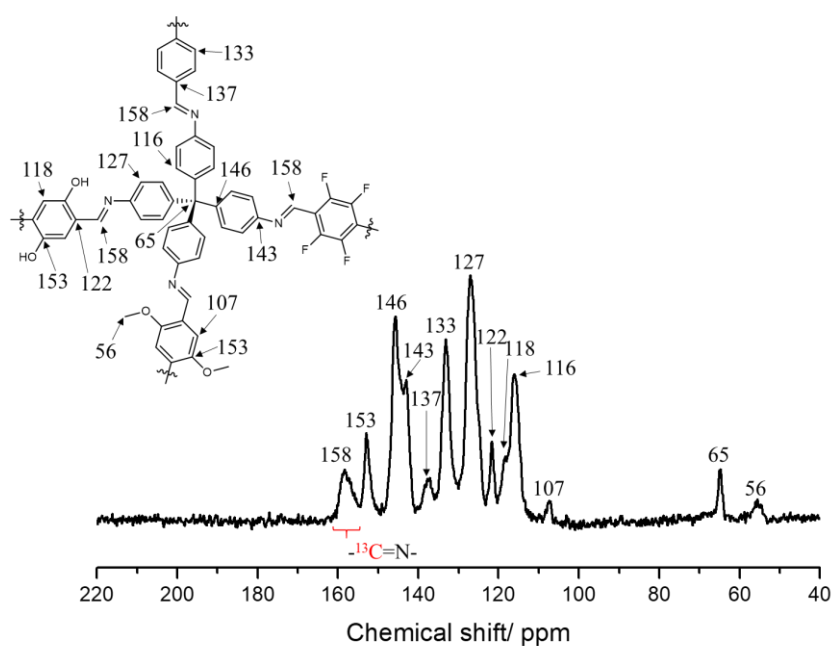
Supplementary Figure 23. ^{13}C CP/MAS spectra of COF-AF, black; knot (TAM), red; linker-A, blue; linker-F, dark cyan.



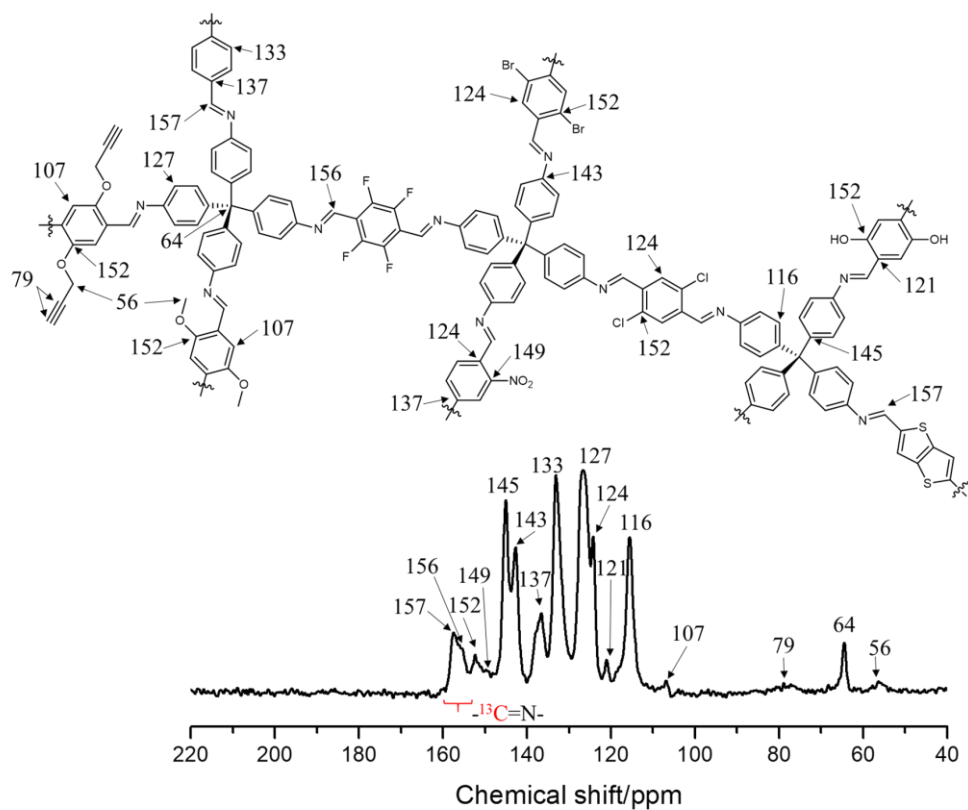
Supplementary Figure 24. ^{13}C CP/MAS spectra of COF-AG, black; knot (TAM), red; linker-A, blue; linker-G, dark cyan.



Supplementary Figure 25. ^{13}C CP/MAS spectra of COF-AH, black; knot (TAM), red; linker-A, blue; linker-H, dark cyan.

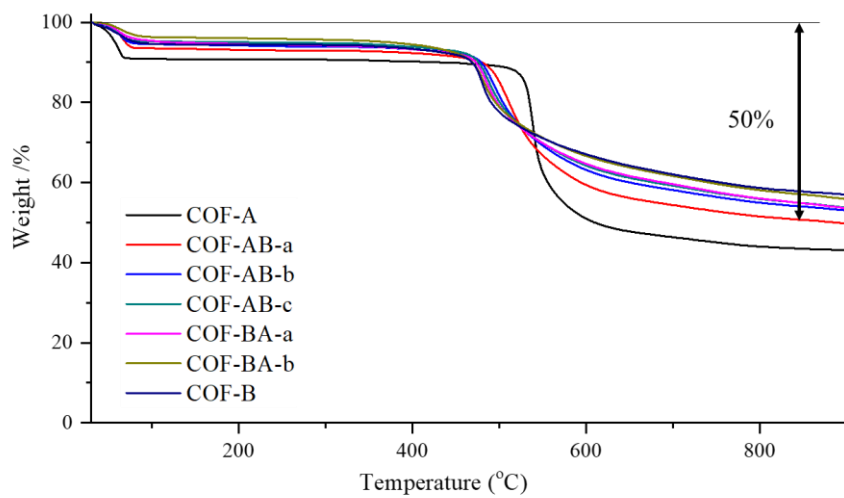


Supplementary Figure 26. ^{13}C CP/MAS spectra of COF-ABDI-a.

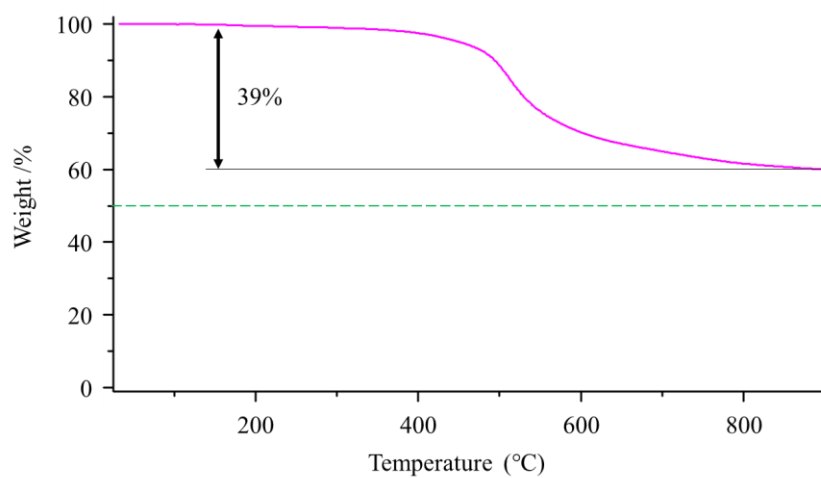


Supplementary Figure 27. ^{13}C CP/MAS NMR spectrum showing the unique resonance of COF-ABCDEFGHI.

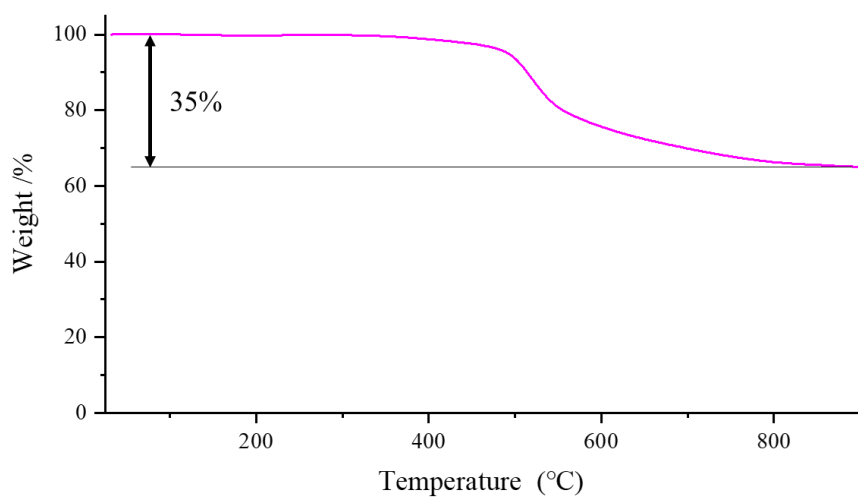
Section 5. Thermogravimetric analysis (TGA)



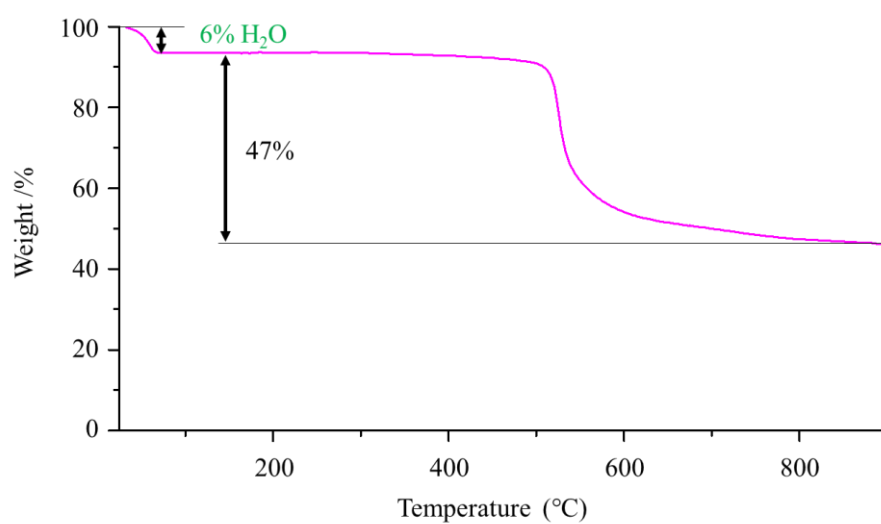
Supplementary Figure 28. TGA trace of COF-ABs and COF-BAs.



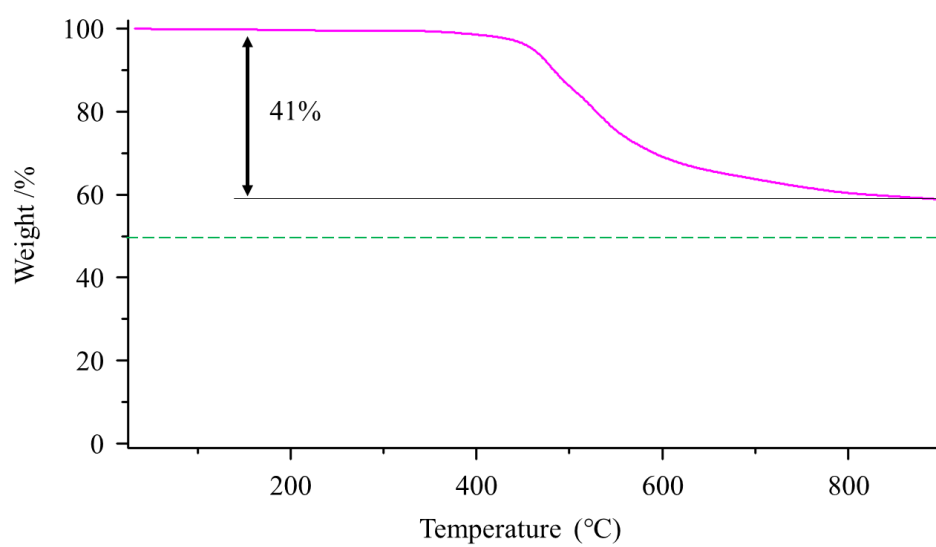
Supplementary Figure 29. TGA trace of COF-AC.



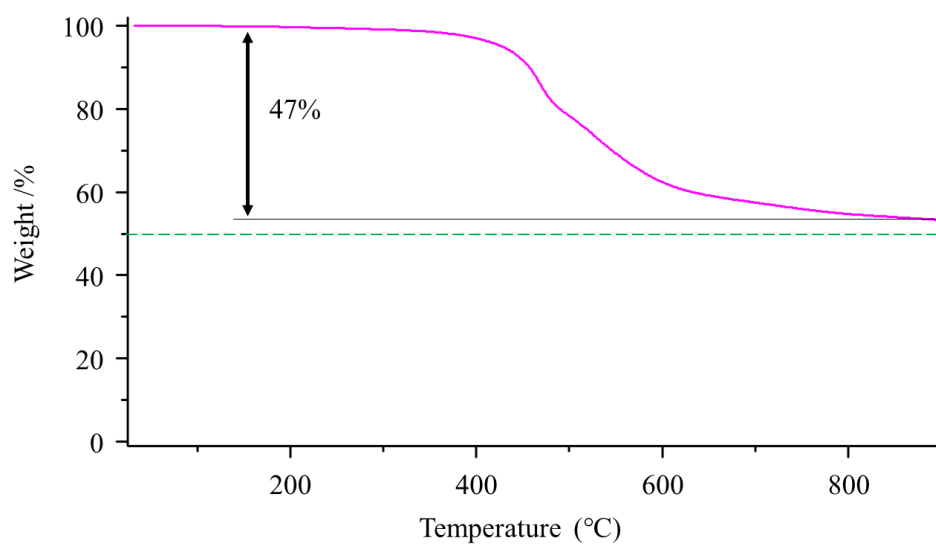
Supplementary Figure 30. TGA trace of COF-AD.



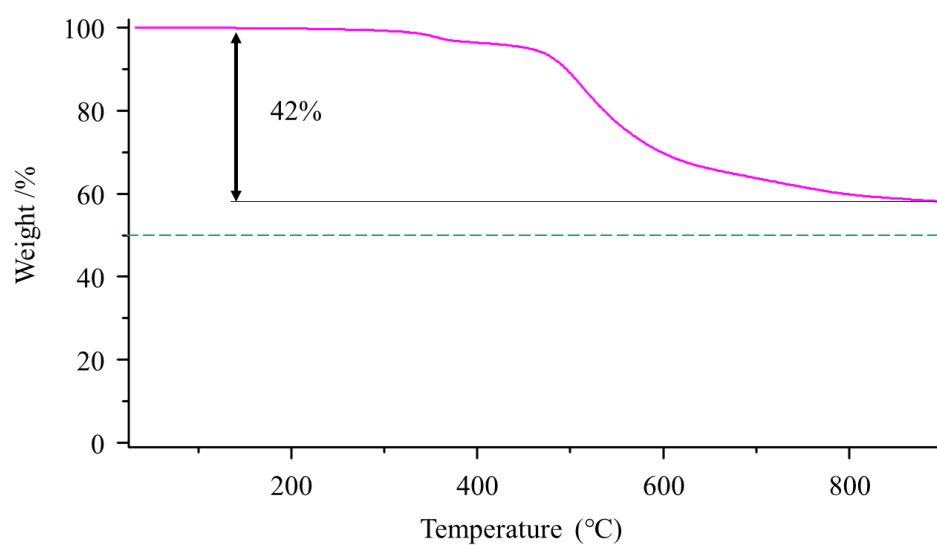
Supplementary Figure 31. TGA trace of COF-AE.



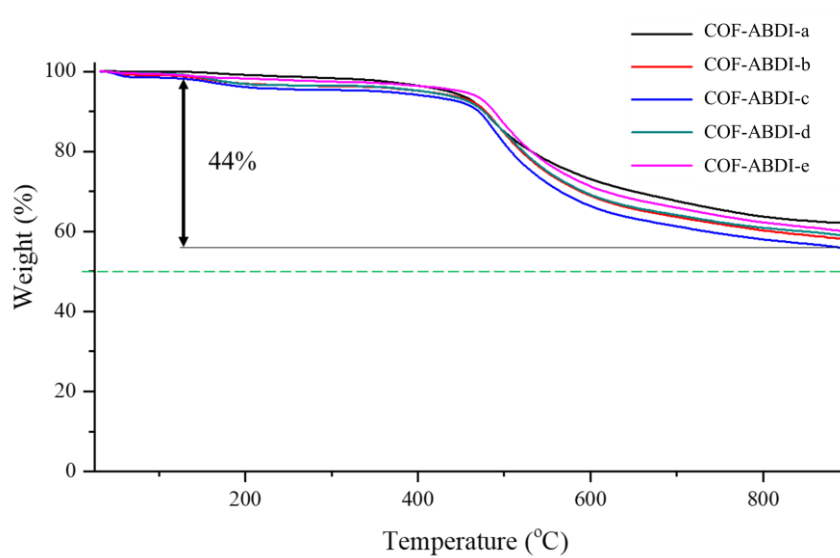
Supplementary Figure 32. TGA trace of COF-AF.



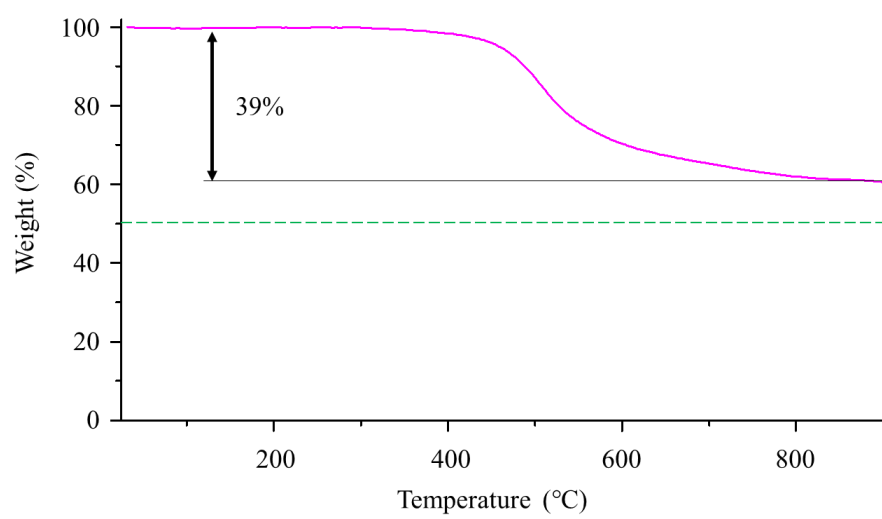
Supplementary Figure 33. TGA trace of COF-AG.



Supplementary Figure 34. TGA trace of COF-AH.



Supplementary Figure 35. TGA trace of COF-ABDIs.



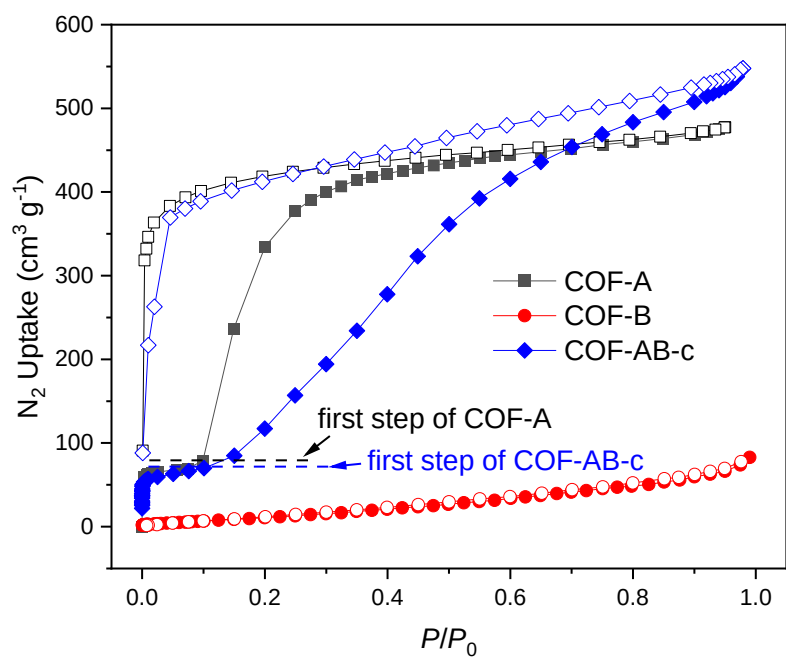
Supplementary Figure 36. TGA trace of COF-ABCDEFGHI.

Section 6. Gas absorption

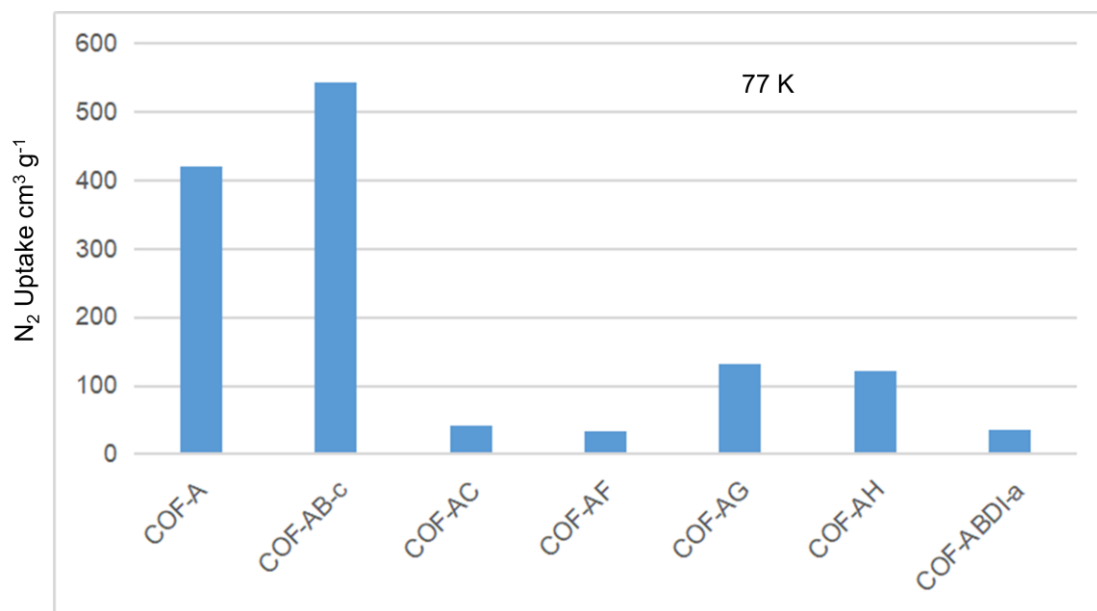
Material Studio was used to simulate the theoretical adsorption capacities of COF-A, COF-B and COF-AB-c, which amounted to $78 \text{ m}^3 \text{ g}^{-1}$, $70 \text{ m}^3 \text{ g}^{-1}$, and $72 \text{ m}^3 \text{ g}^{-1}$, respectively. The experimental adsorption capacity of COF-B is $67 \text{ m}^3 \text{ g}^{-1}$, which is similar to the calculated value. The first step adsorption capacities of both COF-A and COF-AB-c obtained experimentally are similar to the calculated values (**Supplementary Table 12** and **Supplementary Figure 37**), and the experimental values at P/P_0 of 0.95 are $476 \text{ m}^3 \text{ g}^{-1}$ and $526 \text{ m}^3 \text{ g}^{-1}$, respectively. The big difference was attributed to the flexibility of the synthesized COFs, whose lattice volume could be expanded to accommodate more gas molecules under higher adsorption pressure. The expansion of the lattice volume will significantly change the adsorption isotherm and made the calculation of reliable BET surface area with current theory difficult. The experimental BET surface areas of COF-A and COF-AB-c were $1320 \text{ m}^2 \text{ g}^{-1}$ and $1230 \text{ m}^2 \text{ g}^{-1}$, respectively.

Supplementary Table 12. Theoretical and experiment N_2 adsorption capacity of COF-A, COF-B and COF-AB-c based on single crystal structure.

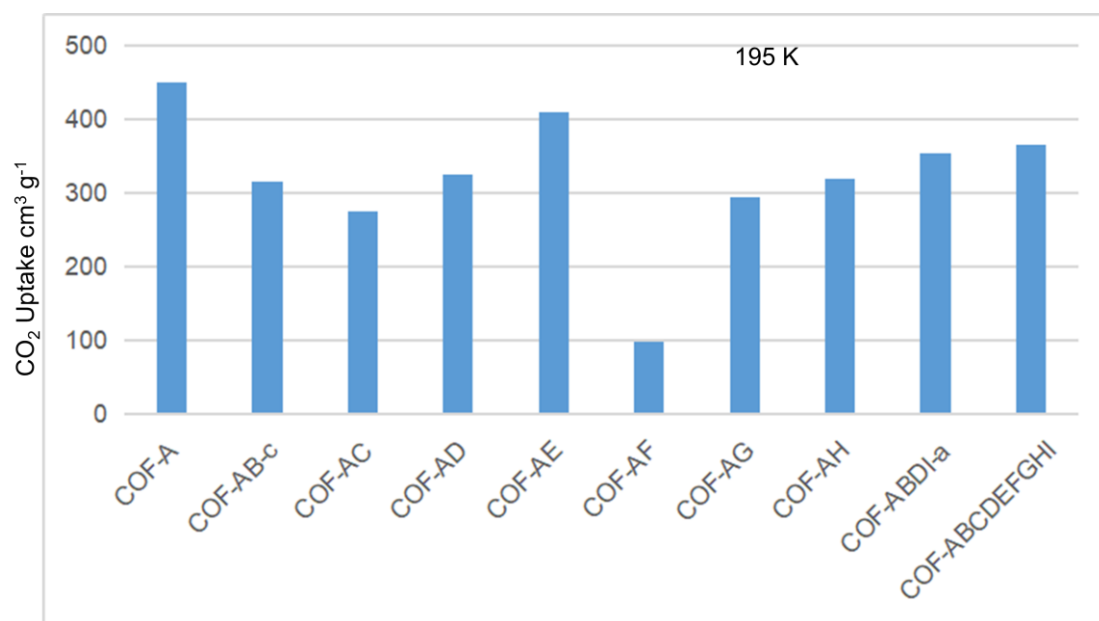
Compounds	Theoretical uptake ($\text{cm}^3 \text{ g}^{-1}$)	Experiment uptake ($\text{cm}^3 \text{ g}^{-1}$)	
		The first step	The second step
COF-A	78	79.6	476
COF-B	70	67	--
COF-AB-c	72	74.2	526



Supplementary Figure 37. N₂ adsorption isotherms of COF-A, COF-B and COF-AB-c at 77K.

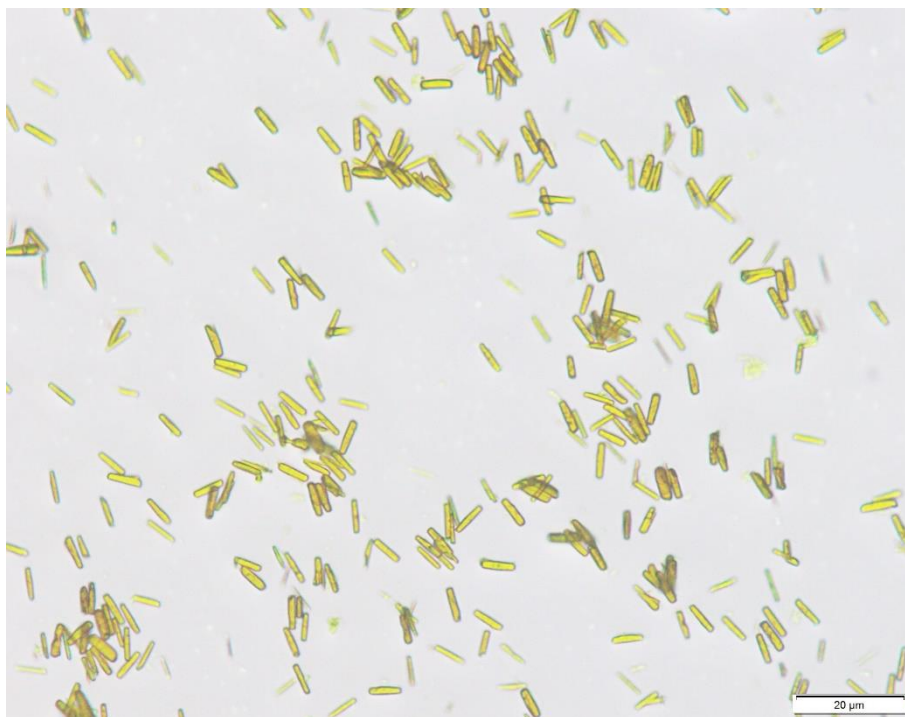


Supplementary Figure 38. N₂ uptake ($P/P_0 = 0.95$) of COF-A, -AB-c, -AC, -AF, -AG, -AH, -ABDI-a at 77K.

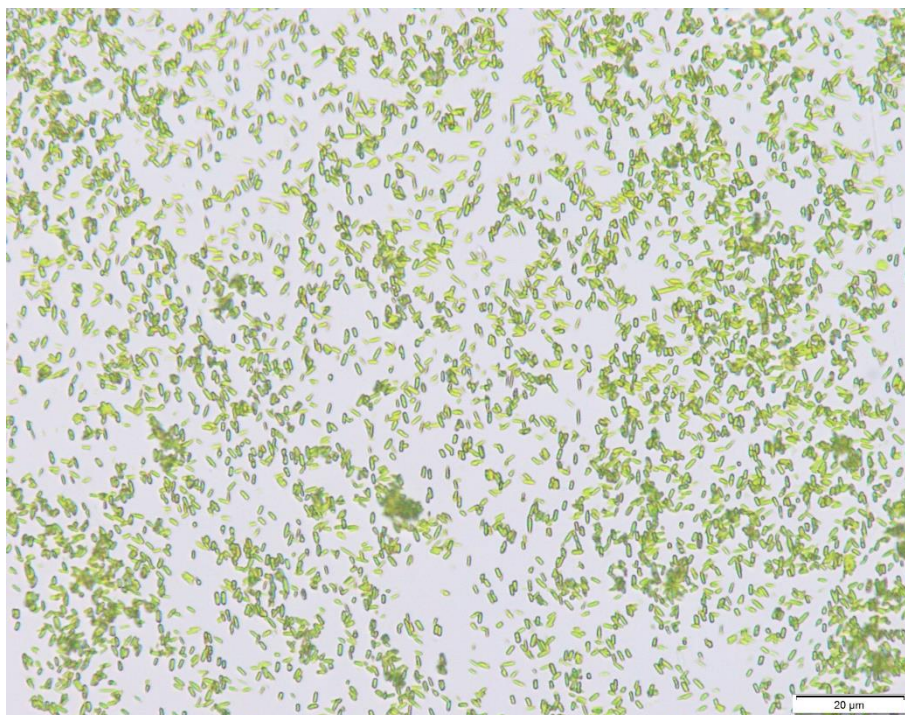


Supplementary Figure 39. CO₂ uptake ($P/P_0 = 0.95$) of COF-A, -AB-c, -AC, -AD, -AE, -AF, -AG, -AH, -ABDI-a and -ABCDEFGHI at 77K.

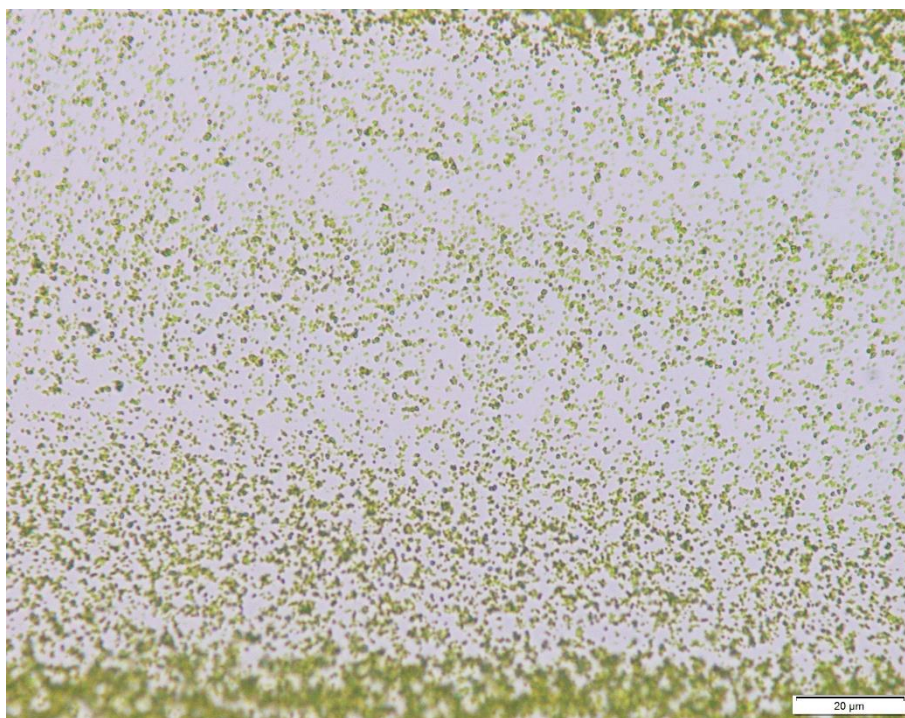
Section 7. Optical microscopic pictures of synthesis COFs



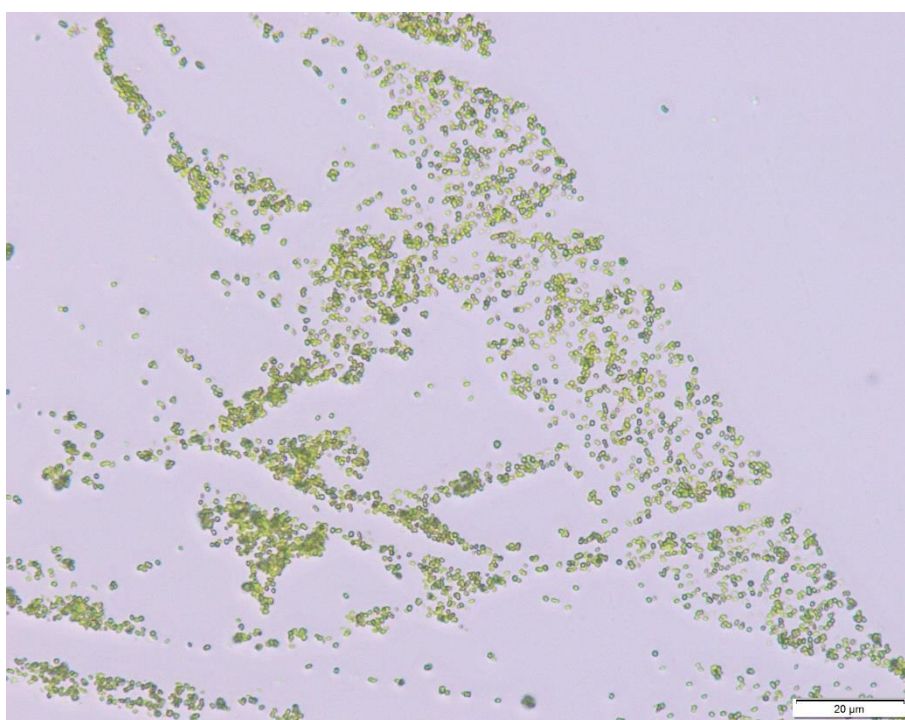
Supplementary Figure 40. Optical microscopic pictures of COF-AB-c



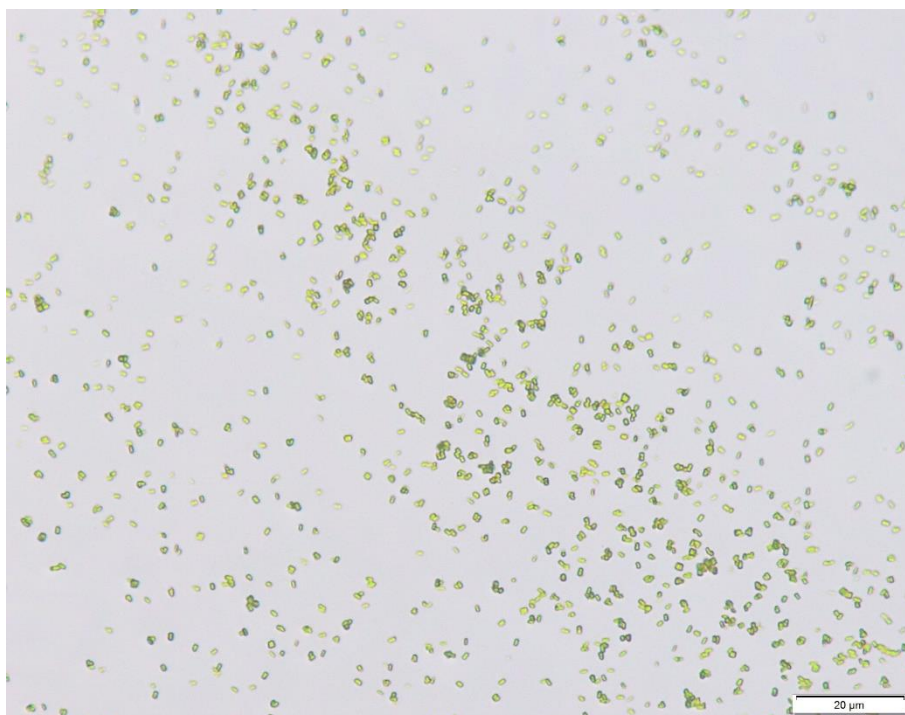
Supplementary Figure 41. Optical microscopic pictures of COF-AC.



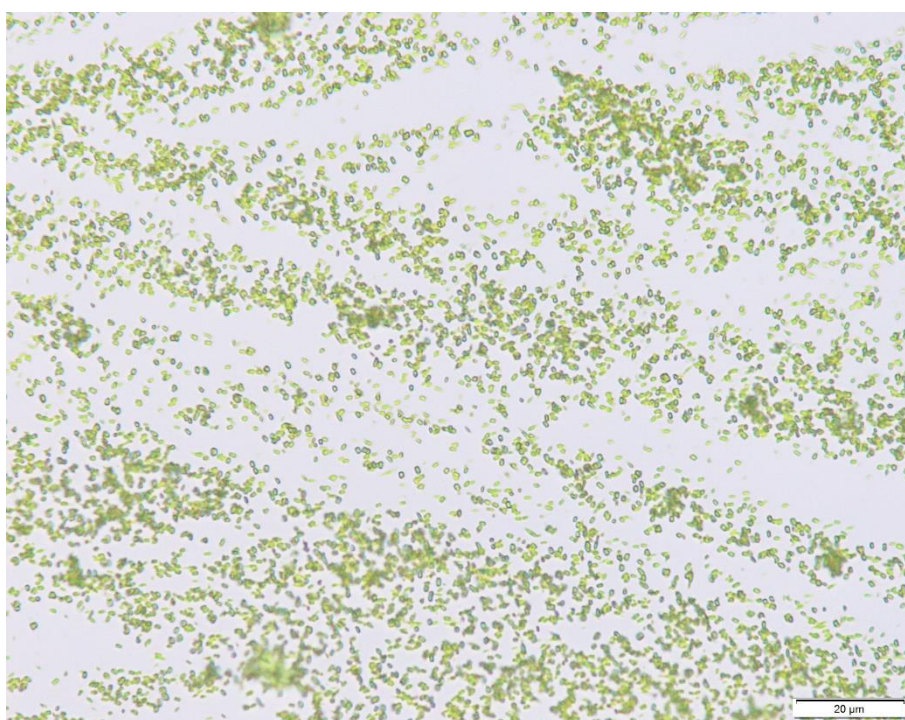
Supplementary Figure 42. Optical microscopic pictures of COF-AD.



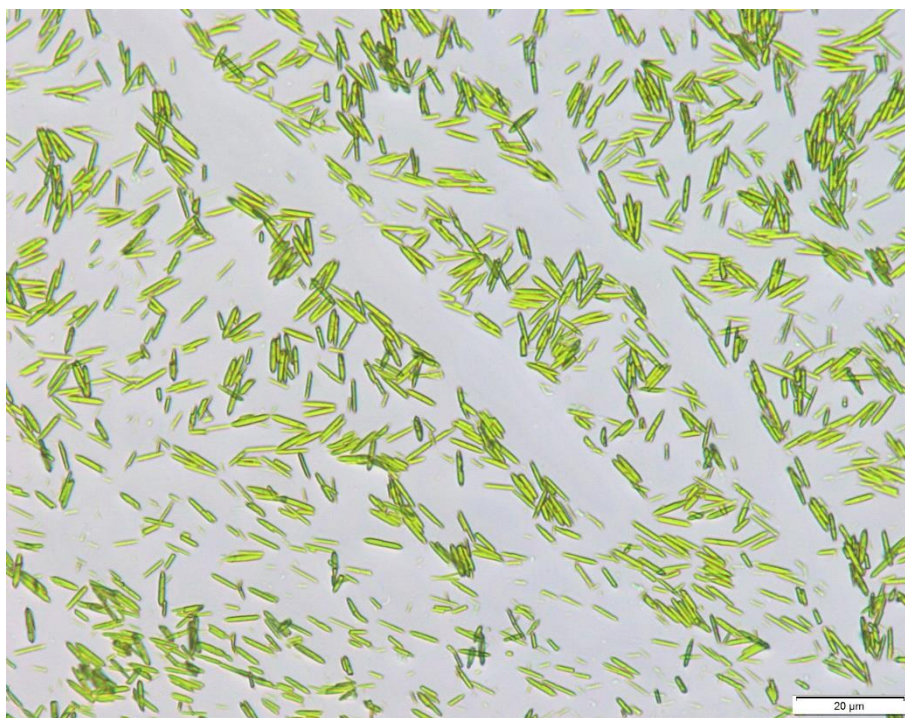
Supplementary Figure 43. Optical microscopic pictures of COF-AE.



Supplementary Figure 44. Optical image of COF-AF.



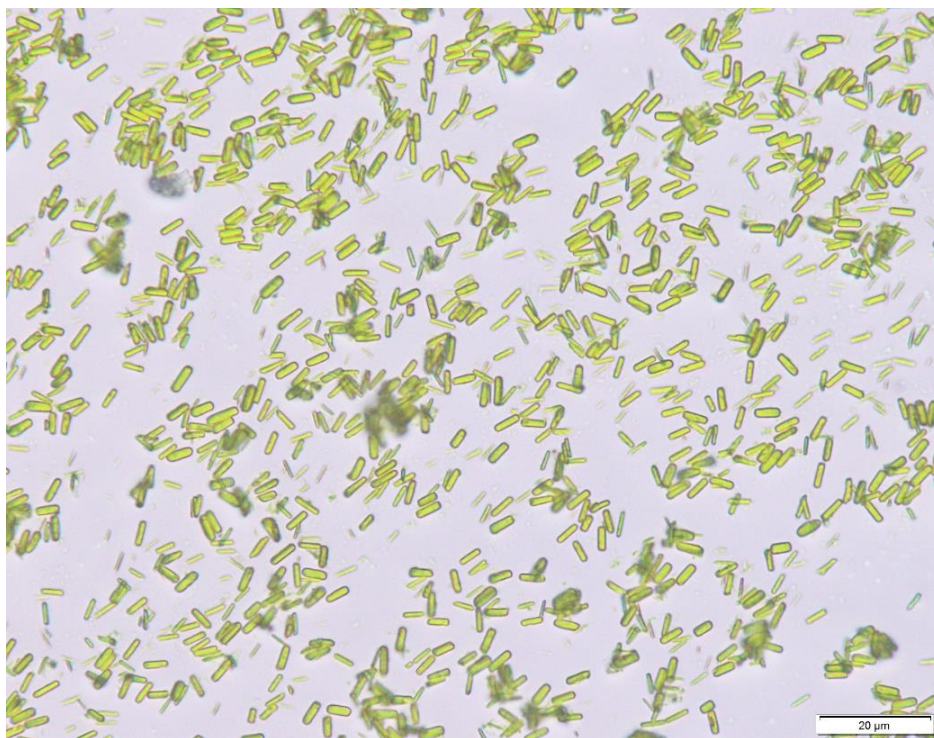
Supplementary Figure 45. Optical image of COF-AG.



Supplementary Figure 46. Optical image of COF-AH.



Supplementary Figure 47. Optical image of COF-ABDI-a.

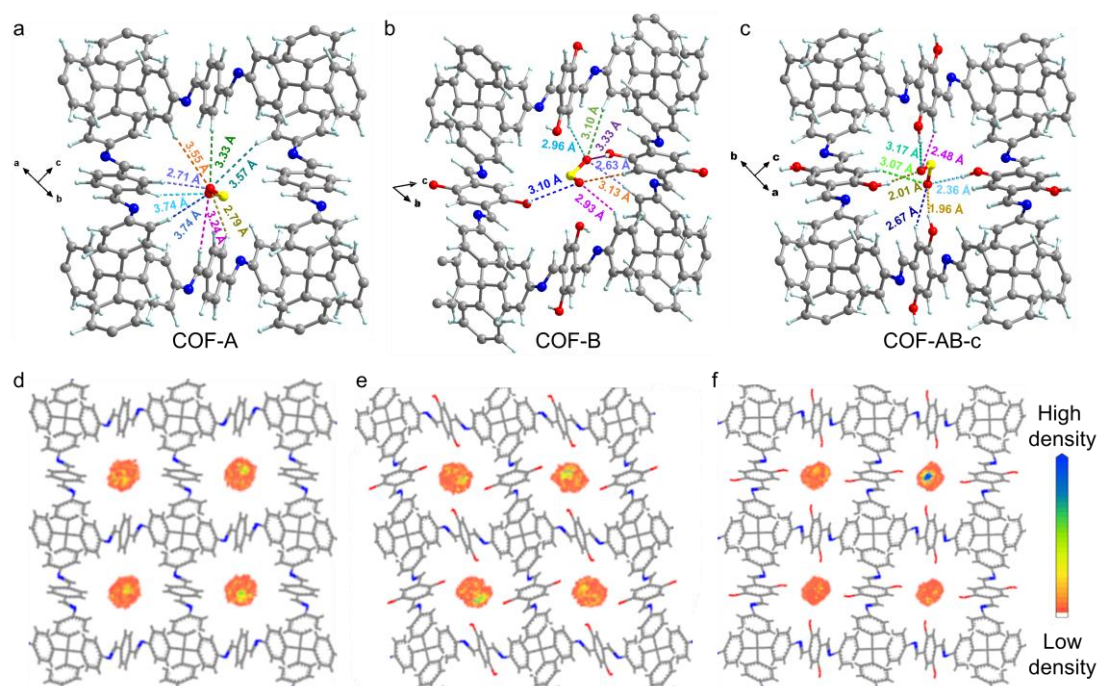


Supplementary Figure 48. Optical image of COF-ABCDEFGHI.

Section 8. Discussion on adsorption mechanism of sulfur dioxide

We then used grand canonical Monte Carlo (GCMC) simulation and first-principle density functional theory (DFT) to simulate the powder XRD patterns upon loading and unloading of SO₂ molecules while keeping the crystal structure of COF-AB-c unchanged. As can be seen from Figure 4c, the (200) peak exhibits a diminishing intensity trend in response to the increase of loaded SO₂ molecules, while the intensity of peaks at higher diffraction angles gradually strengthens. In contrast, the intensity of diffraction peaks returned to their initial state after removal of adsorbed SO₂ molecules within the pore channels (Figure 4b). As can be seen from **Supplementary Figure 49 a, b, c** SO₂ molecule can be favorably anchored in the pores by multiple O···H–O hydrogen bonds and O···H–C hydrogen bonds. The pore of COF-A forms eight-fold hydrogen interactions with SO₂ molecule, and the strongest hydrogen bonds come from the O···H interaction with the hydrogen atoms at the periphery of benzene ring on the pore wall (bond length: 2.71/2.79 Å). COF-B exhibits a diamond shaped channel, SO₂ molecule forms seven-fold hydrogen bonding interactions (bond length: ~ 2.63 – 3.33 Å) with the hydrogen atoms and hydroxyl groups at the periphery of benzene rings on the pore wall. The strongest interaction is provided by hydroxyl group of linker-B (bond length: 2.63 Å). With pores of COF-AB-c, seven hydrogen bonds (1.96~3.17 Å) were identified, the strongest three hydrogen bonds (bond strength: 1.96, 2.01 and 2.36 Å) interactions are all contributed by –OH groups, these interactions are significantly stronger than those between COF-A and COF-B. The adsorption density distributions of SO₂ within pores of COF-A, COF-B and COF-AB-c have also been analyzed by grand canonical Monte Carlo (GCMC) calculations which indicate the favorable adsorption locations of SO₂ are all included in the high density region, consistent with the favorable locations of SO₂ within the pores of COF-A, COF-B and COF-AB-c calculated by DFT (**Supplementary Figure 49 d, e, f**). Moreover, the DFT calculated SO₂ isosteric heat (Q_{st}) for COF-AB-c (191.98 kJ mol⁻¹) is higher than that of COF-A (143.75 kJ mol⁻¹) and COF-B (152.55

kJ mol^{-1}), further confirms the SO_2 adsorption trend unveiled by the experimental measurements (Supplementary Table 13).



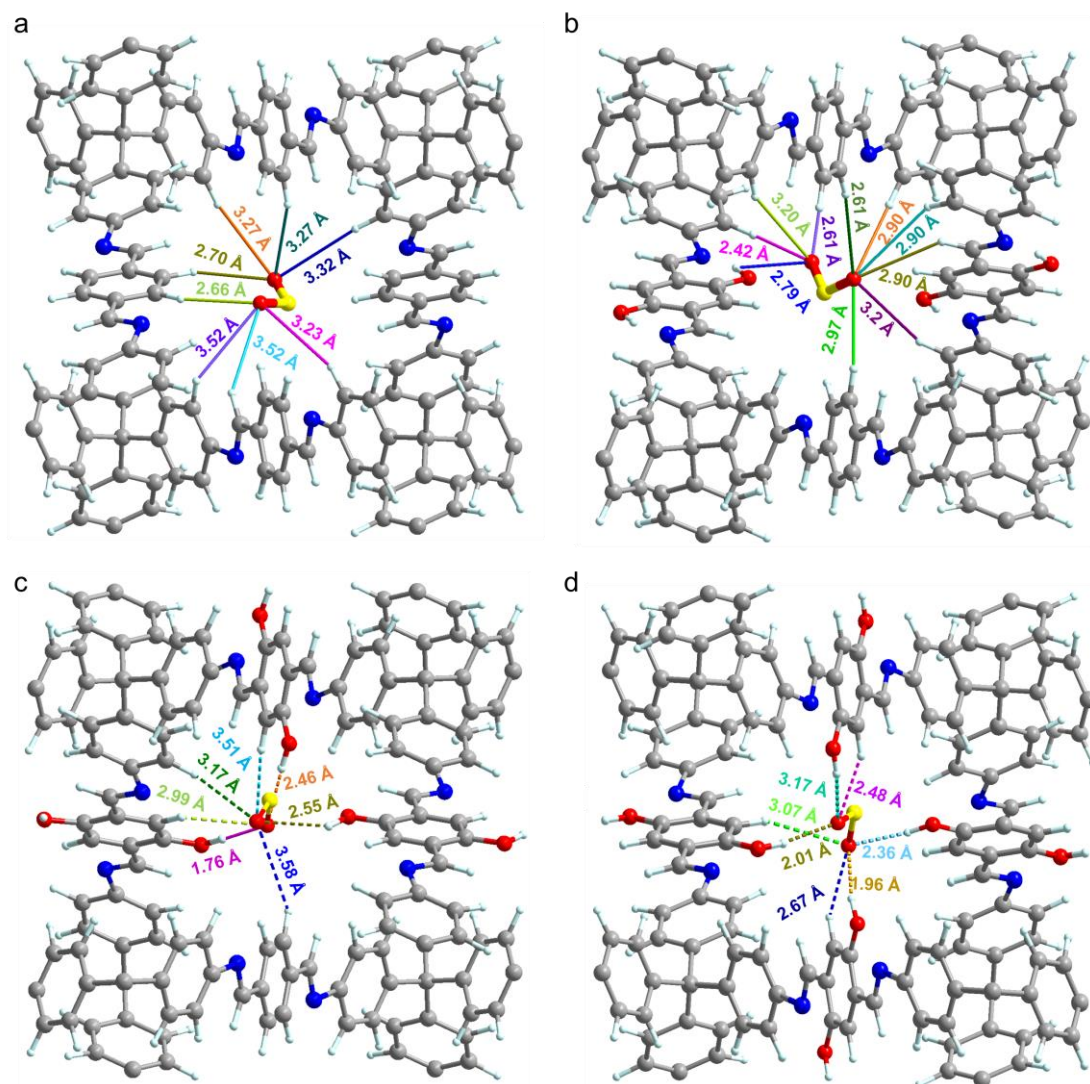
Supplementary Figure 49. Adsorption sites of SO_2 in optimized COF-A (a), COF-B (b) and COF-AB-c (c) calculated by density function theory. Density distribution of SO_2 in optimized COF-A (d) COF-B (e) and COF-AB-c (f) at 298K and 100 kPa.

Supplementary Table 13. Summary of DFT calculation results of COF-A, COF-B and COF-AB with different sequences.

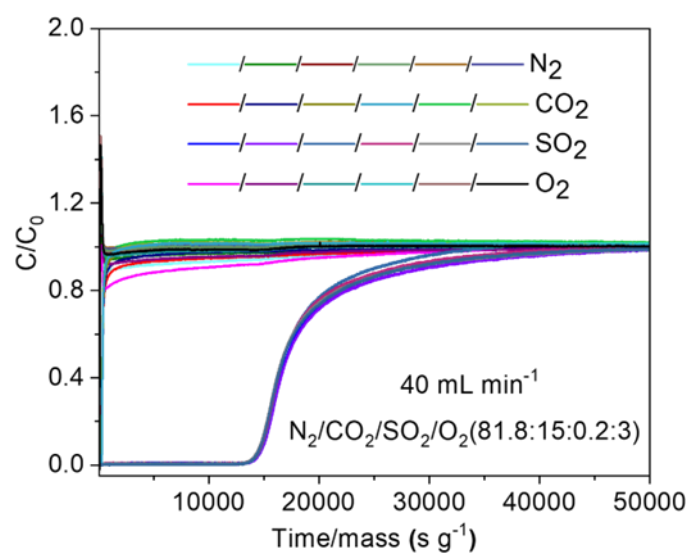
Adsorbent	COF-AB-AAAA	COF-AB-ABAB	COF-AB-ABBB	COF-AB-BBBB	COF-AB-BBBB	COF-AB-BBBB
Q_{st}						
(kJ mmol^{-1})	147.85	154.22	178.81	199.49	143.75	152.55

Section 9. Discussion of the effect of sequence on SO₂ adsorption

To prove the assumption, we used density function theory to investigate the interactions between SO₂ and COF-AB-cs with four sequenced arrangement of linker-A and linker-B, namely COF-AB-AAAA (the last part of the name of the crystals represents the linker and their sequence at the pore wall of the lattice), COF-AB-ABAB, COF-AB-ABBB, and COF-AB-BBBB. These results shows SO₂ molecule adsorbed in the pore channel forms hydrogen bonds with the hydrogen atoms and hydroxyl groups at the periphery of benzene rings on the pore wall (**Supplementary Figure 50**). The location of the SO₂ molecule shifts or rotates depending on the sequence of the linkers since the hydrogen bonds with hydroxyl groups are stronger than those with hydrogen atoms. Both the location and sequence of the hydroxyl groups affects the hydrogen interactions and therefore the adsorption of SO₂. We also calculated the isosteric heat for the adsorption of SO₂ with the COF-ABs, COF-A and COF-B to estimate the interactions between and the SO₂ pores of the COFs. The value isosteric heat increase upon the increase of hydroxyl groups in the sequence, and that of COF-AB-AAAA and COF-AB-ABAB is higher than that of COF-A and COF-B, respectively (**Supplementary Table 13**). In comparison with pure component COFs, different components can form various sequence to tune the locations of SO₂ and its interaction strength within the pores, and therefore tune the adsorption and diffusion of SO₂ as well as other gas, leading to optimized uptake capacity and selectivity in the end.



Supplementary Figure 50. DFT calculated adsorption sites of SO_2 in optimized COF-AB-AAAA (a), COF-AB-ABAB (b), COF-AB-ABBB (c) and COF-AB-BBBB (d).



Supplementary Figure 51. Six cycling column breakthrough tests for COF-AB-c.

Supplementary References

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checkCIF/PLATON report

Structure factors have been supplied for datablock(s) COF-AB-c

THIS REPORT IS FOR GUIDANCE ONLY. IF USED AS PART OF A REVIEW

PROCEDURE FOR

PUBLICATION, IT SHOULD NOT REPLACE THE EXPERTISE OF AN EXPERIENCED

CRYSTALLOGRAPHIC REFEREE.

No syntax errors found. CIF dictionary Interpreting this report

Datablock: COF-AB-c

Bond precision:

C-C = 0.0112 Å

Wavelength=0.02508

Cell:

a=19.92000

b=19.92000

c=8.64000

alpha=90

beta=90

gamma=90

Temperature: 100 K

Calculated

Reported

Volume

3428.407

3428

Space group

I 41/a:1 I 41/a

Hall group

I 4bw -1bw -I 4ad

Moiety formula

C164 H104.74 N16 O7.26,

12.64(O)

?

Sum formula

C164 H104.74 N16 O19.90

C41 H28 N4 O1.814

Mr

2617.72

605.70

Dx,g cm⁻³

1.268

1.173

Z

1
4
Mu (mm⁻¹)
0.000
0.000
F000
0.0
1360.0
F000'
1359.68
h, k, lmax
20, 22, 9
Nref
1133
Tmin, Tmax
Tmin'
Correction method= Not given
Data completeness= Theta(max)= 0.800
R(reflections)= 0.3239(441)
wR2(reflections)=
wR= 0.4438(1133)
S = 3.670
Npar= 58The following ALERTS were generated. Each ALERT has the format
test-name_ALERT_alert-type_alert-level.
Click on the hyperlinks for more details of the test.

Alert level A
ATOM007_ALERT_1_A _atom_site_aniso_label is missing
Unique label identifying the atom site.
Author Response: The structure was refined isotropically.
PLAT082_ALERT_2_A High R1 Value 0.32 Report

Author Response: The structure is not refined anisotropically which causes high R values And the multiple scattering of electrons in the crystal also lead to a higher R
PLAT201_ALERT_2_A Isotropic non-H Atoms in Main Residue(s) 12 Report
N1 C1 C2 C3 C4 C5 C6 C7
C8 C9 C10 C11
Author Response: In this case, reliable ADPs could not be refined. This is acommon issue for electron diffraction measurements.
PLAT410_ALERT_2_A Short Intra H...H Contact H1C4 ..H1C8 . 1.76 Ang.
x, y, z = 1_555 Check

Author Response: Refinement is only based on kinematical assumption.

checkCIF/PLATON report

Structure factors have been supplied for datablock(s) COF-ABDI-a

THIS REPORT IS FOR GUIDANCE ONLY. IF USED AS PART OF A REVIEW

PROCEDURE FOR

PUBLICATION, IT SHOULD NOT REPLACE THE EXPERTISE OF AN EXPERIENCED

CRYSTALLOGRAPHIC REFEREE.

No syntax errors found. CIF dictionary Interpreting this report

Datablock: COF-ABDI-a

Bond precision:

C-C = 0.0059 Å

Wavelength=0.02508

Cell:

a=19.34000

b=19.34000

c=8.74000

alpha=90

beta=90

gamma=90

Temperature: 100 K

Calculated

Reported

Volume

3269.071

3269

Space group

I 41/a:1 I 41/a

Hall group

I 4bw -1bw -I 4ad

Moiety formula

C41 H23.95 F2.40 N4 O1.65,

2.16(O)

?

Sum formula

C41 H23.95 F2.40 N4 O3.81

C41.39 H26.38 F2.4 N4 O1.65

Mr

679.15

651.70

Dx,g cm⁻³

1.380

1.324

Z

4
4
Mu (mm⁻¹)
0.000
0.000
F000
0.0
1401.0
F000'
1399.88
h, k, lmax
23, 25, 11
Nref
2028
Tmin, Tmax
Tmin'
Correction method= Not given
Data completeness= Theta(max)= 0.960
R(reflections)= 0.2386(1260)
wR2(reflections)=
wR= 0.2973(2028)
S = 3.720
Npar= 68The following ALERTS were generated. Each ALERT has the format
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Click on the hyperlinks for more details of the test.

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Unique label identifying the atom site.
Author Response: The structure was refined isotropically.
PLAT082_ALERT_2_A High R1 Value 0.24 Report

Author Response: The structure is not refined anisotropically which causes high R values And the multiple scattering of electrons in the crystal also lead to a higher R
PLAT201_ALERT_2_A Isotropic non-H Atoms in Main Residue(s) 12 Report
N1 C1 C2 C3 C4 C5 C6 C7
C8 C9 C10 C11
Author Response: In this case, reliable ADPs could not be refined. This is a common issue for electron diffraction measurements.

checkCIF/PLATON report

Structure factors have been supplied for datablock(s) COF-ABCDEFGHI
THIS REPORT IS FOR GUIDANCE ONLY. IF USED AS PART OF A REVIEW
PROCEDURE FOR
PUBLICATION, IT SHOULD NOT REPLACE THE EXPERTISE OF AN EXPERIENCED
CRYSTALLOGRAPHIC REFEREE.

No syntax errors found. CIF dictionary Interpreting this report

Datablock: COF-ABCDEFGHI

Bond precision:

C-C = 0.0071 Å

Wavelength=0.02508

Cell:

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b=19.95000

c=8.67000

alpha=90

beta=90

gamma=90

Temperature: 100 K

Calculated

Reported

Volume

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3451

Space group

I 41/a:1 I 41/a

Hall group

I 4bw -1bw -I 4ad

Moiety formula

C164 H80 N16 O11.20

?

Sum formula

C164 H80 N16 O11.20

C41 N4 O2.797 H20

Mr

2453.64

613.39

Dx, g cm⁻³

1.181

1.181

Z

1

4
 Mu (mm-1)
 0.000
 0.000
 F000
 0.0
 1265.0
 F000'
 1265.37
 h, k, lmax
 24, 19, 10
 Nref
 1703
 Tmin, Tmax
 Tmin'
 Correction method= Not given
 Data completeness= Theta(max)= 0.900
 R(reflections)= 0.2486(746)
 wR2(reflections)=
 wR= 0.3279(1703)
 S = 4.420
 Npar= 51The following ALERTS were generated. Each ALERT has the format
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 Click on the hyperlinks for more details of the test.
Alert level A
 ATOM007_ALERT_1_A _atom_site_aniso_label is missing
 Unique label identifying the atom site.
Author Response: The structure was refined isotropically.
 PLAT082_ALERT_2_A High R1 Value 0.25 Report
Author Response: The structure is not refined anisotropically which causes high R values And the multiple scattering of electrons in the crystal also lead to a higher R
 PLAT201_ALERT_2_A Isotropic non-H Atoms in Main Residue(s) 13 Report
 O1 N1 C1 C2 C3 C4 C5 C6
 C7 C8 C9 C10 C11
Author Response: In this case, reliable ADPs could not be refined. This is a common issue for electron diffraction measurements.